

Dynamic Scheduling - Scoreboard -

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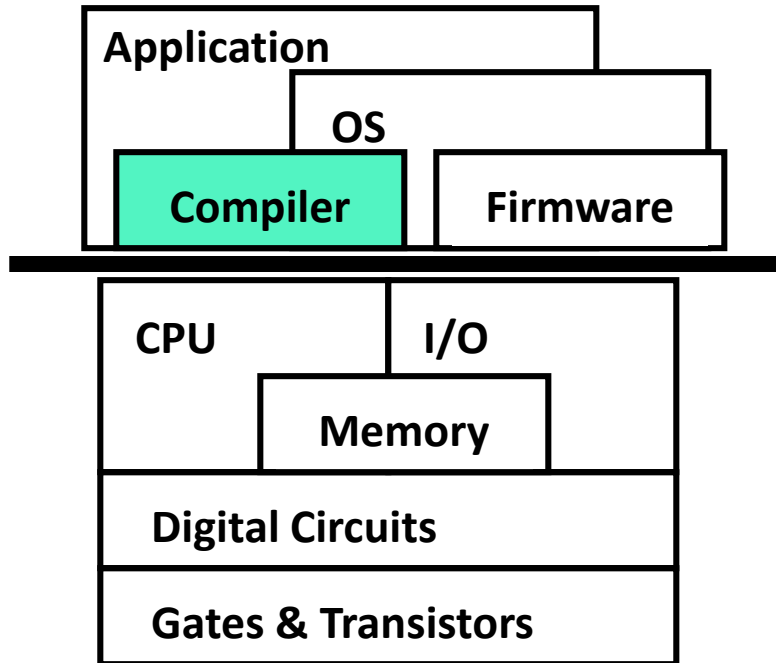
KAIST EE

CAMELab



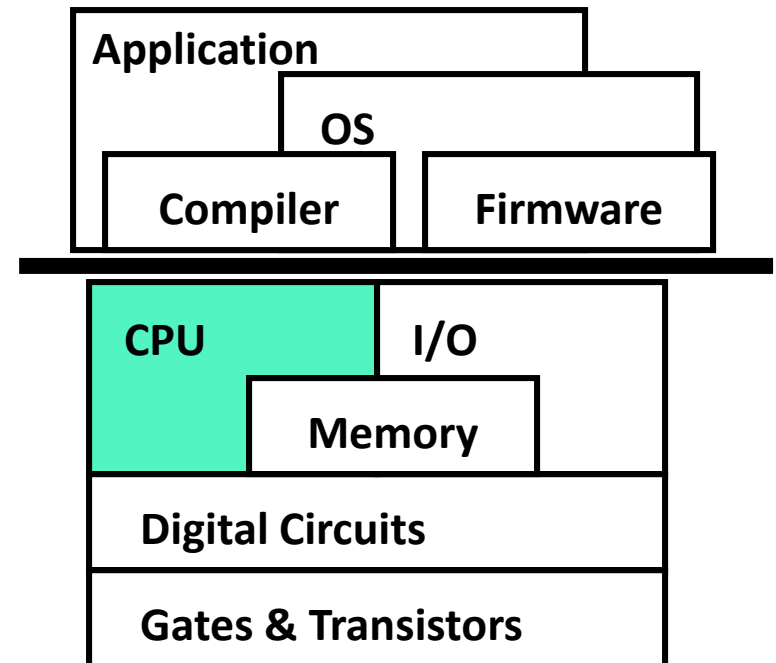
Recall: Instruction Scheduling

Static Scheduling



- Conventional techniques
 - Loop unrolling
- Advanced ILP-target techniques
 - Global scheduling & Software pipelining

Dynamic Scheduling



- Scoreboard
- Tomaulo's algorithm
- Reorder buffer (ROB)

Dynamic Scheduling means Out-of-Order execution (OoO; O3)

Let's think about the following questions

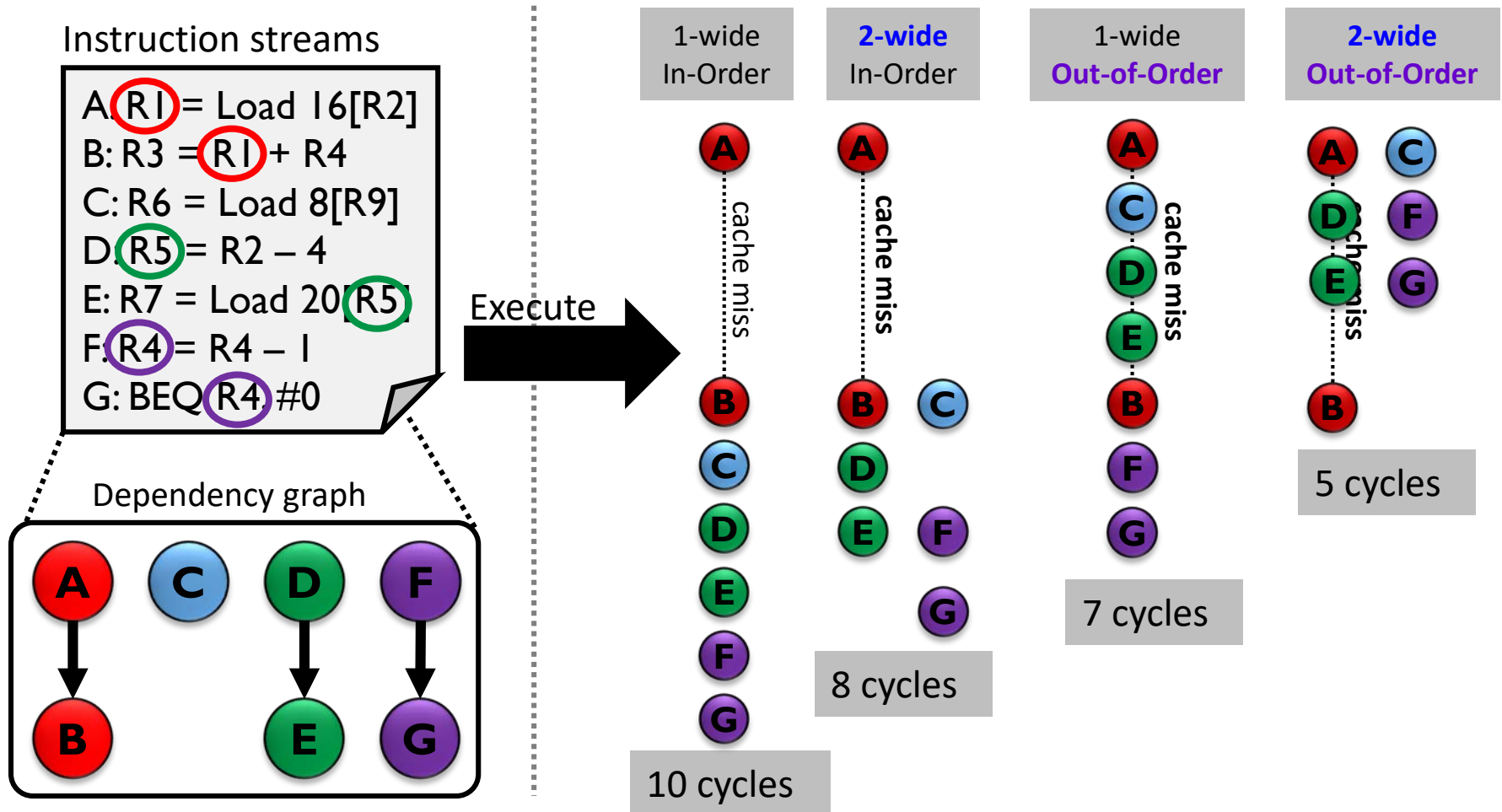
Q1) How can the O3 achieve **performance benefits**?

Q2) Is everything okay? Any **problems** of O3?

Q3) How does the O3 **work**?

Q1) How can O3 achieve perf. benefits?

➤ **Hardware** rearranges instruction stream to reduce stalls

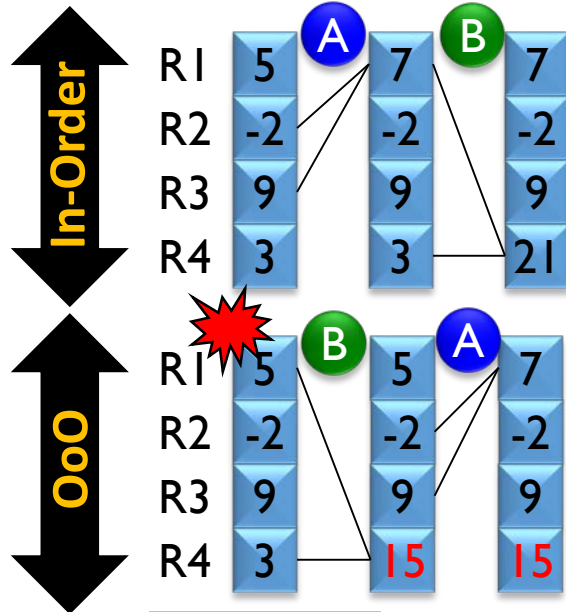


Q2) Any problems of O3?

➤ Recall: **Hazards!** Especially for register dependencies

Read-After-Write

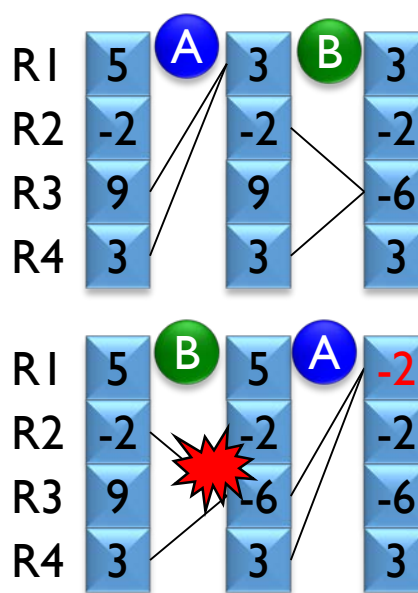
A: $R1 = R2 + R3$
B: $R4 = R1 * R4$



Read
old data

Write-After-Read

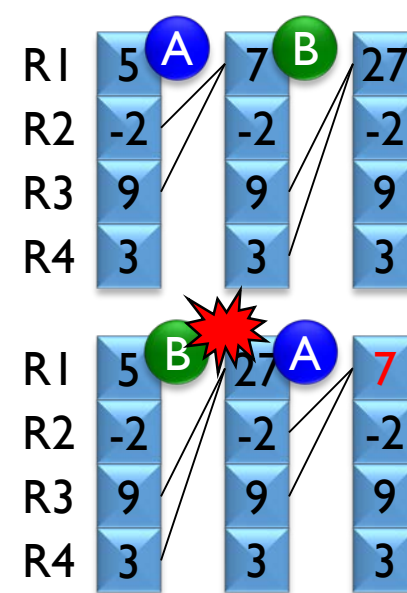
A: $R1 = R3 / R4$
B: $R3 = R2 * R4$



Read future
data

Write-After-Write

A: $R1 = R2 + R3$
B: $R1 = R3 * R4$



Will be overwritten
by legacy data

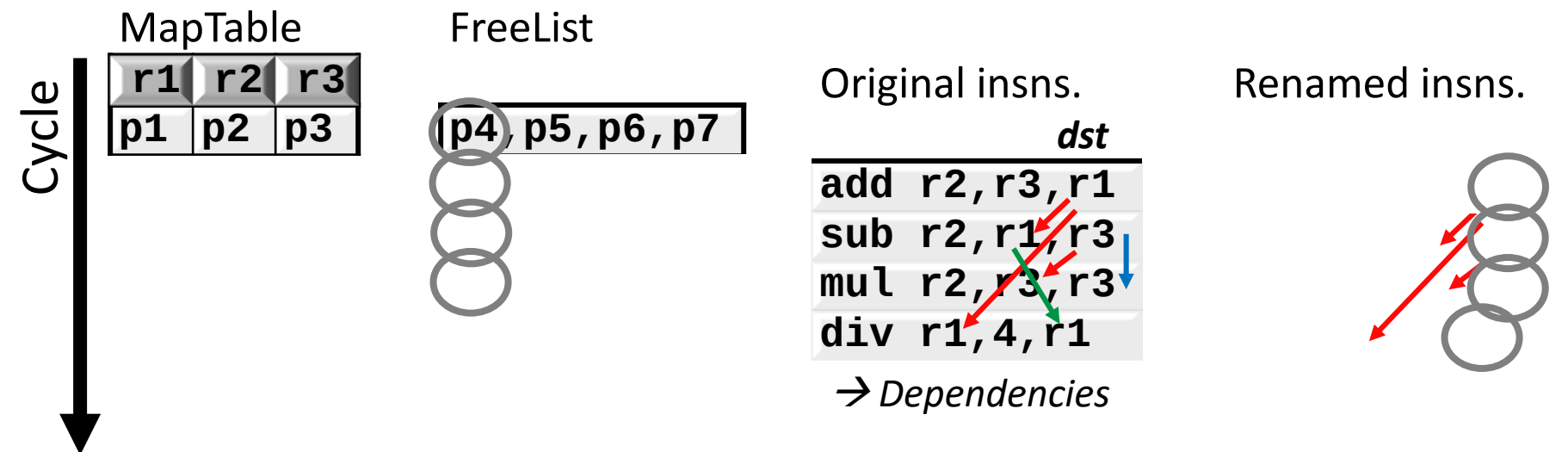
→ Cycle

Recall: Register Renaming

- Solution for **WAR**, **WAW** hazards; done in hardware
 - Dependency is on name/location, not on data
 - Renaming removes WAR/WAW, but leaves **RAW** intact

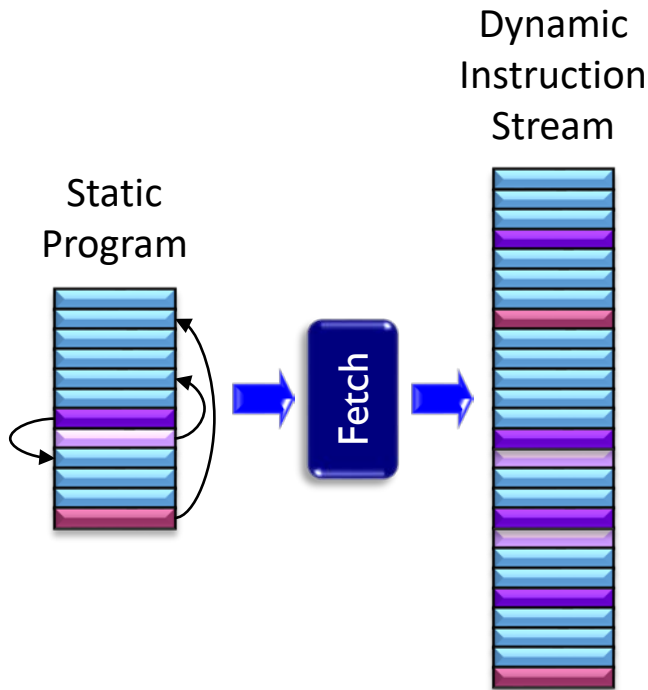
Q) How does the register renaming work?

A) Allocate new location by noting in map-table



Q3) How does the O3 work?

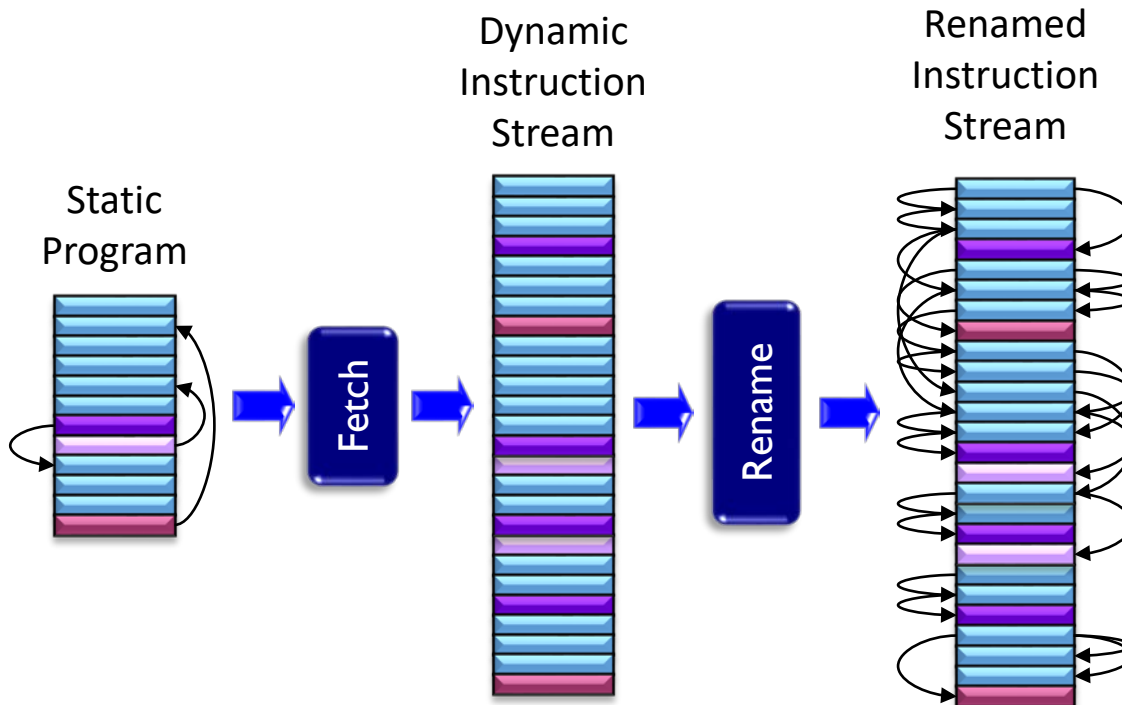
- **Step1:** Fetch many instructions into *instruction window*



Today's machines: 100+ instructions scheduling window
Use branch prediction to speculate past branches (will be covered in later)

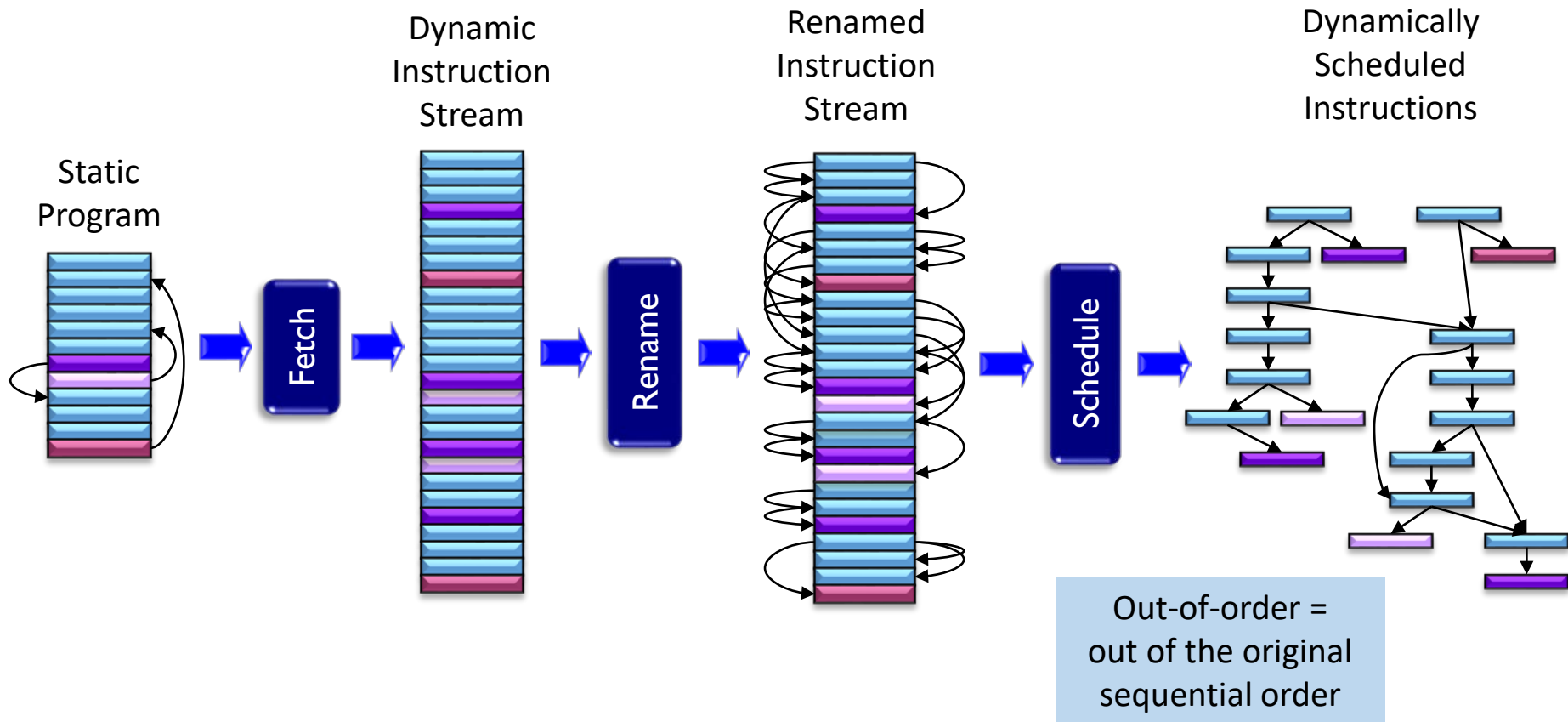
Q3) How does the O3 work?

➤ **Step2:** Rename regs. to avoid false deps. (WAW and WAR)

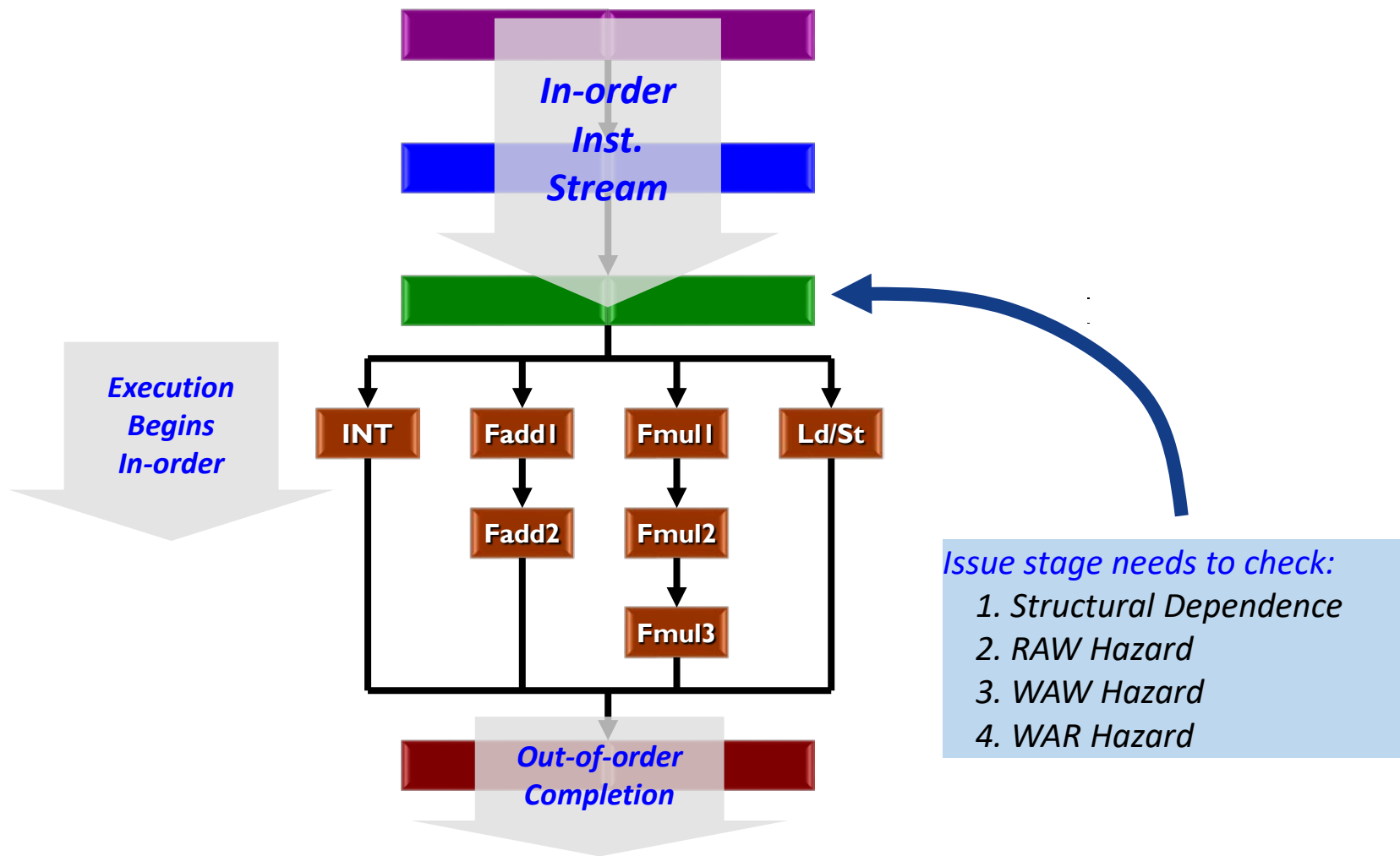


Q3) How does the O3 work?

- **Step3:** Execute instructions as soon as dependencies (registers and memory) are known



➤ **Note that O3 means Out-of-Order Completion**
In-Order Issue; Instructions are issued in order!





Dynamic Scheduling I: Scoreboard

- Let's track the flow of the instrs, register, and function units
 - to check which Datapath components are using / can be used
 - to find out which instruction could be executed without hazards

The CDC 6600 Project ['1964]

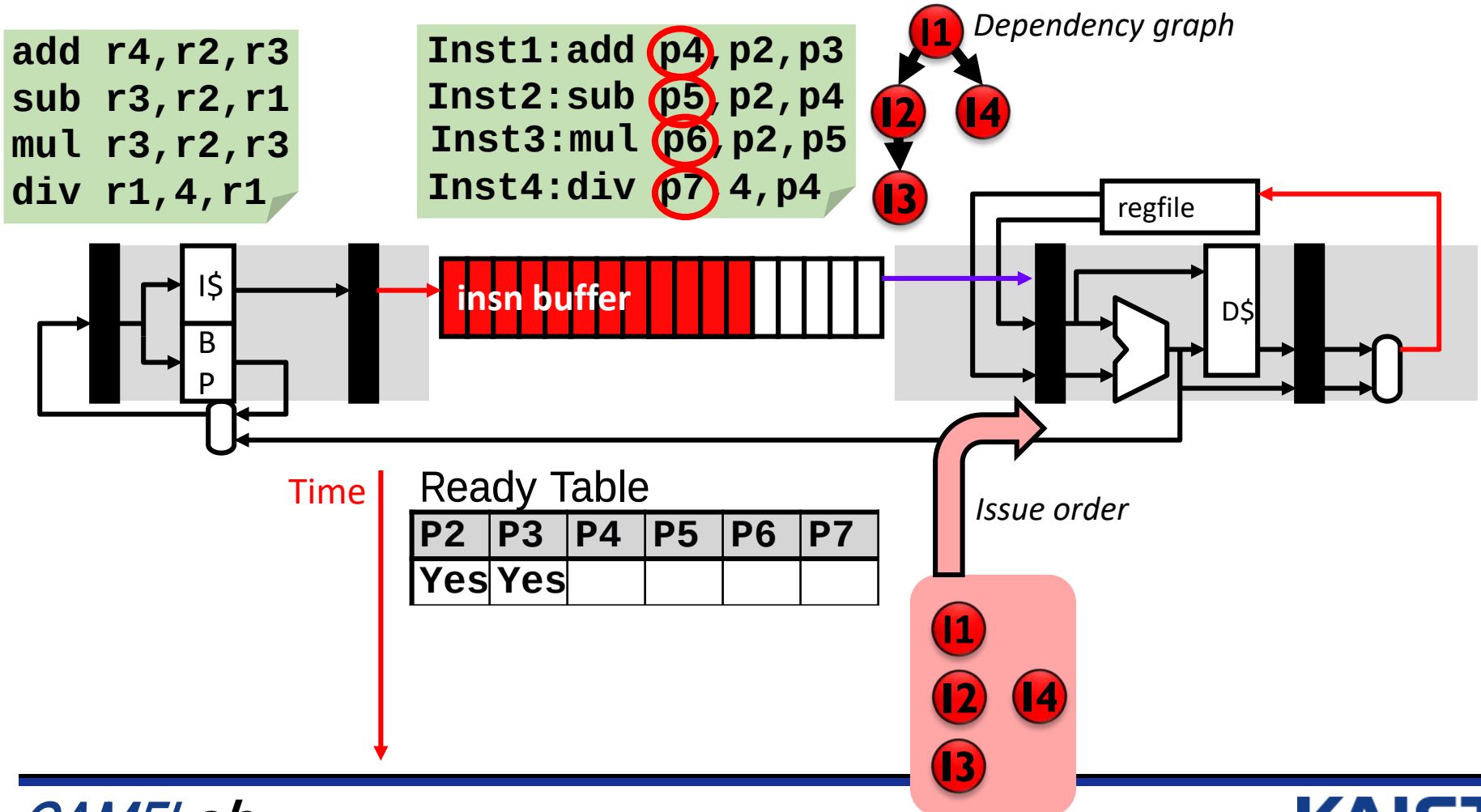
First implementation of
Scoreboard

- 16 separate non-pipelined functional units (7 int, 4FP, 5 memory)
- No register by passing



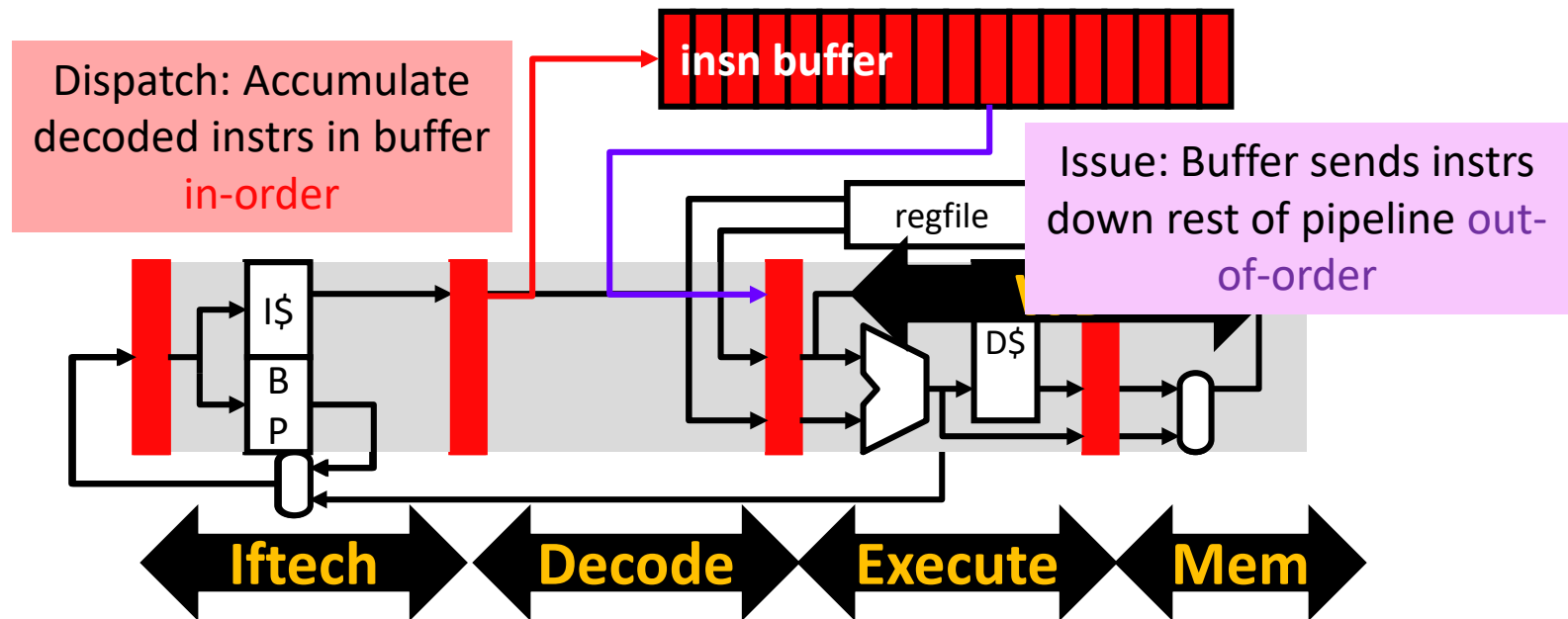
Dynamic Scheduling: The Big Picture

- Instructions fetch/decoded/renamed into *Instruction Buffer*
- Instructions (conceptually) check ready bits every cycle



Stage Splitting w/ Instruction Buffer

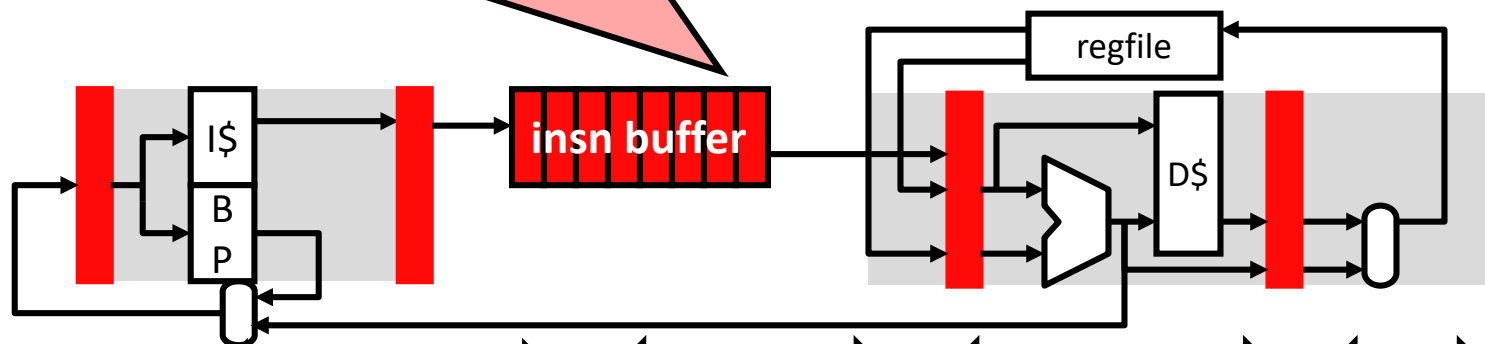
The key idea of Scoreboard



- Conventional 5 pipeline stages
Iftech → Decode → Execute → Memory → Writeback
- Let's split Decode stage into two sub stages
Dispatch (D) & Issue (S)

Scoreboard Pipeline

A new kind of structural hazard
: Instruction buffer is **full**



Dispatch(D)

Issue(S)

Excute (Ex)

WB

Allocate slot in
instruction buffer
and dispatch
instruction **in-order**

Solve structural
hazard & WAW!
Check FU and
destination reg

Multiple
Functional
units

Solve WAR!
Stall until earlier
instruction reads
the destination
register

Simple Scoreboard Data Structures

- Three main components
 - Instruction status
 - Functional unit status
 - Register result status

Scoreboard

Insn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2		FU
LD	F6	34+	R2					Int										F0	
LD	F2	45+	R3					Mult1										F2	
MULTD	F0	F2	F4					Mult2										F4	
SUBD	F8	F6	F2					Add										F6	
DIVD	F10	F0	F6					Div										F8	
ADD	F6	F8	F2															F10	
																		...	

► Instruction Status

- Which of 4 steps the instruction is in
 - D: Issue
 - S: Read operands
 - X: Execute stage completion
 - W: Write result to registers

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2				
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int									
Mult1									
Mult2									
Add									
Div									

Reg Status	
	FU
F0	
F2	
F4	
F6	
F8	
F10	
...	

Functional Unit Status

- Indicates the state of the functional unit (FU)
- 9 fields for each FU

B: Indicates whether the unit is busy or not

Op: Operation to perform in the unit (e.g., + or -)

dst: Destination register

src1,src2: Source-register numbers

Q1, Q2: Functional units producing source registers src1, src2

R1, R2: Flags being set when src1/src2 is ready

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2				
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int									
Mult1									
Mult2									
Add									
Div									

Reg Status	
	FU
F0	
F2	
F4	
F6	
F8	
F10	
...	

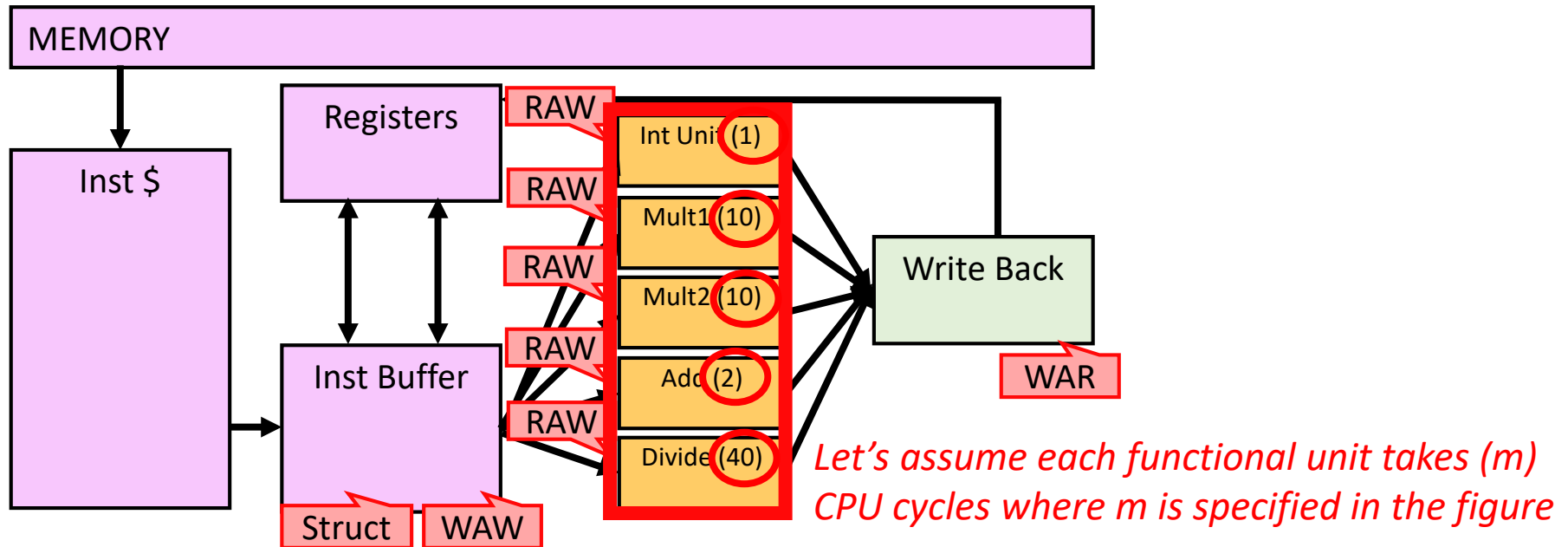
Register Result Status

- Indicates which functional unit will **write** each register, if one exists.
- Blank when no pending instructions will write that register

Scoreboard

Insn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2		FU
LD	F6	34+	R2					Int										F0	
LD	F2	45+	R3					Mult1										F2	
MULTD	F0	F2	F4					Mult2										F4	
SUBD	F8	F6	F2					Add										F6	
DIVD	F10	F0	F6					Div										F8	
ADD	F6	F8	F2															F10	
																		...	

Our Example: “Simple Scoreboard”



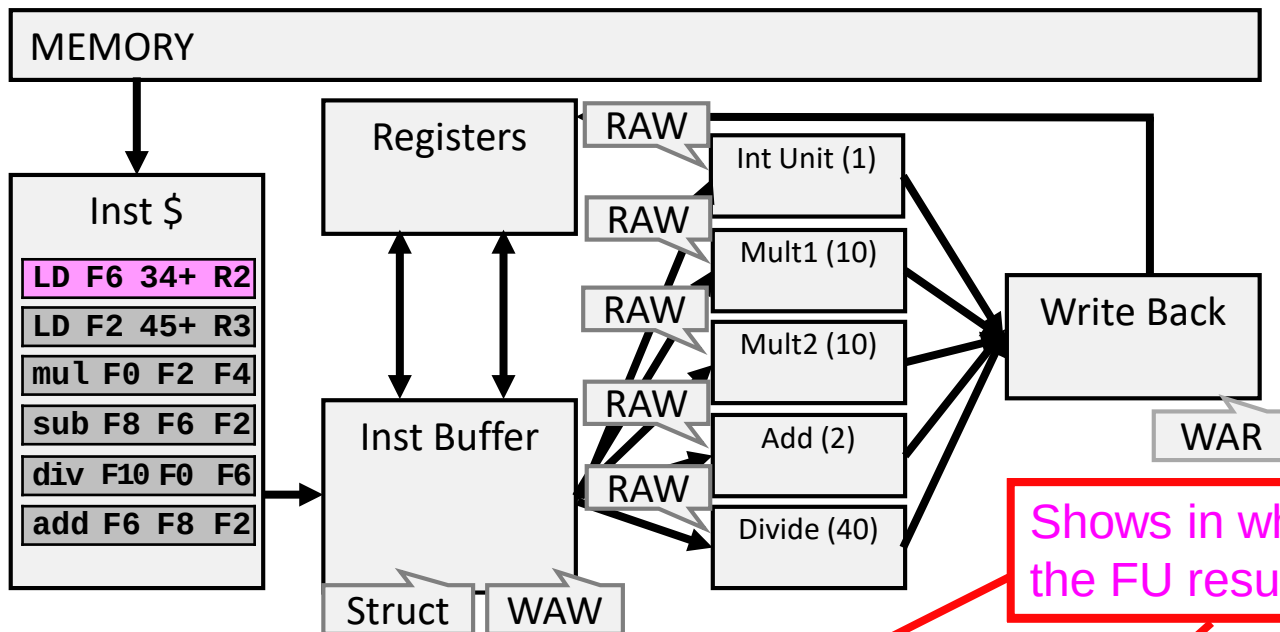
Scoreboard

Instn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2	Reg	FU
LD	F6	34+	R2					Int										F0	
LD	F2	45+	R3					Mult1										F2	
MULTD	F0	F2	F4					Mult2										F4	
SUBD	F8	F6	F2					Add										F6	
DIVD	F10	F0	F6					Div										F8	
ADD	F6	F8	F2															F10	
																		...	

Scoreboard Example: Cycle 1

Try to Issue LD #1

Check Struct & WAW Hazards
→ No Hazards!



Shows in which register the FU result is updated

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1			
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F6		R2				yes
Mult1	No								
Mult2	No								
Add	No								
Div	No								

Reg Status	
	FU
F0	
F2	
F4	
F6	Int
F8	
F10	
...	

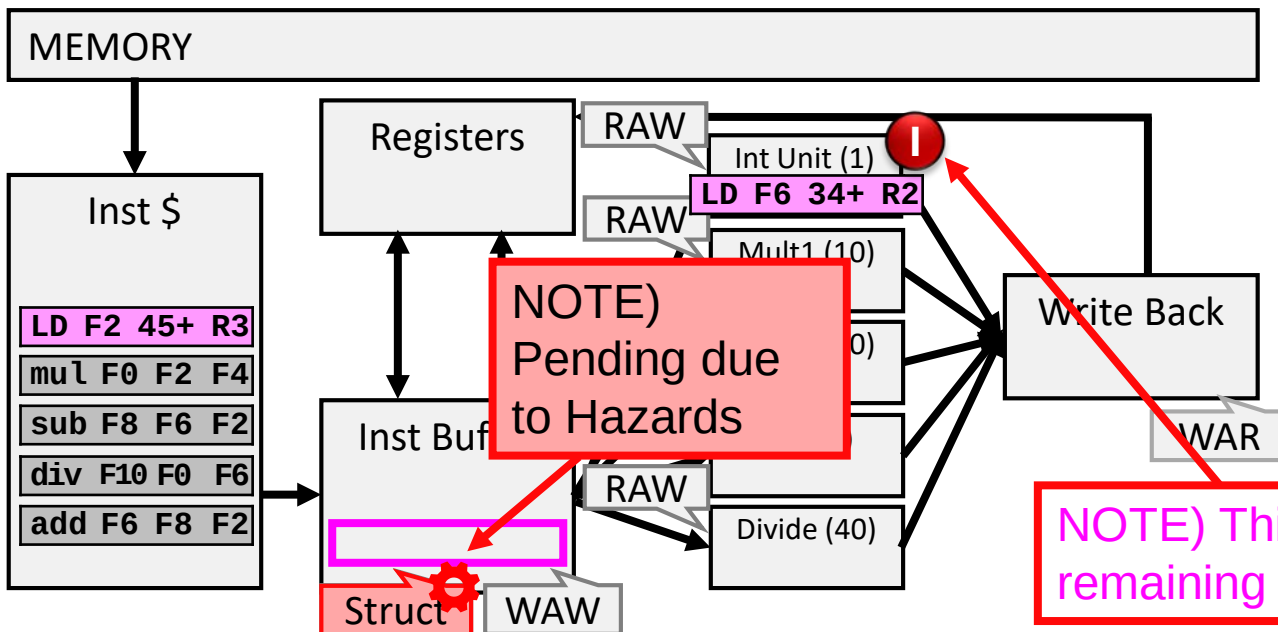
Scoreboard Example: Cycle 2

Read operand
of LD#1

LD #2 can't issue
since int unit is busy.
MULT can't issue
because we require
in-order issue.

NOTE) This is
remaining time

NOTE)
Pending due
to Hazards



Scoreboard

Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2		
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

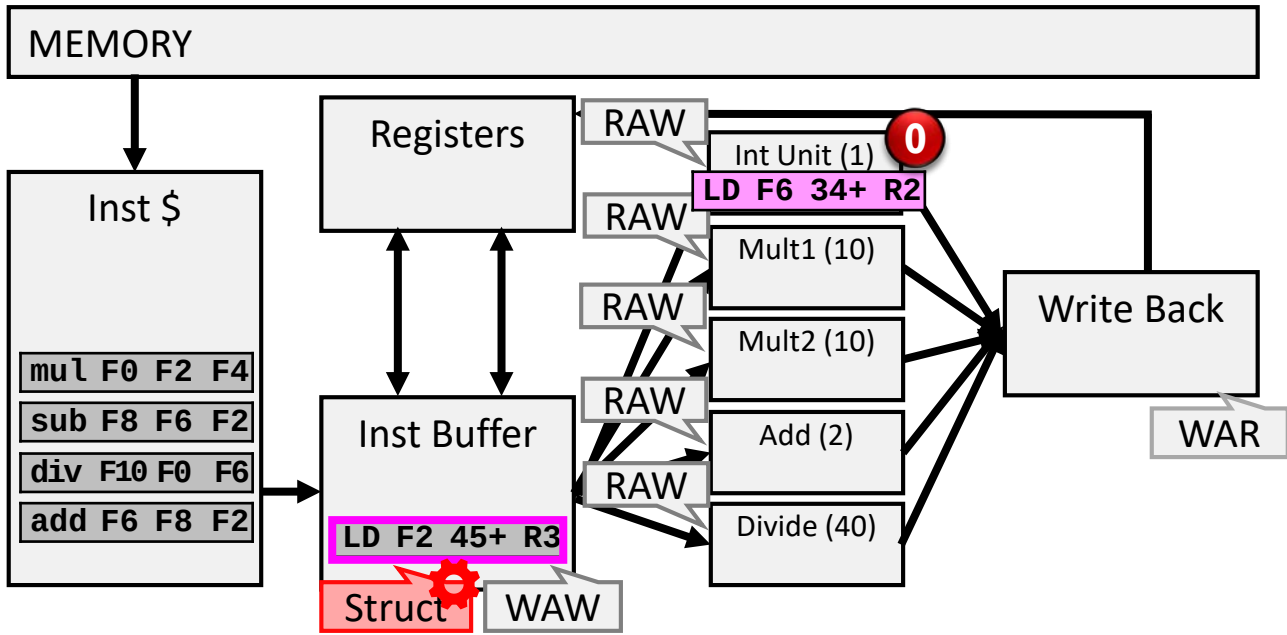
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F6		R2				yes
Mult1	No								
Mult2	No								
Add	No								
Div	No								

Reg Status	
	FU
F0	
F2	
F4	
F6	Int
F8	
F10	
...	

Scoreboard Example: Cycle 3

Execute LD#1

LD#1 comp. done



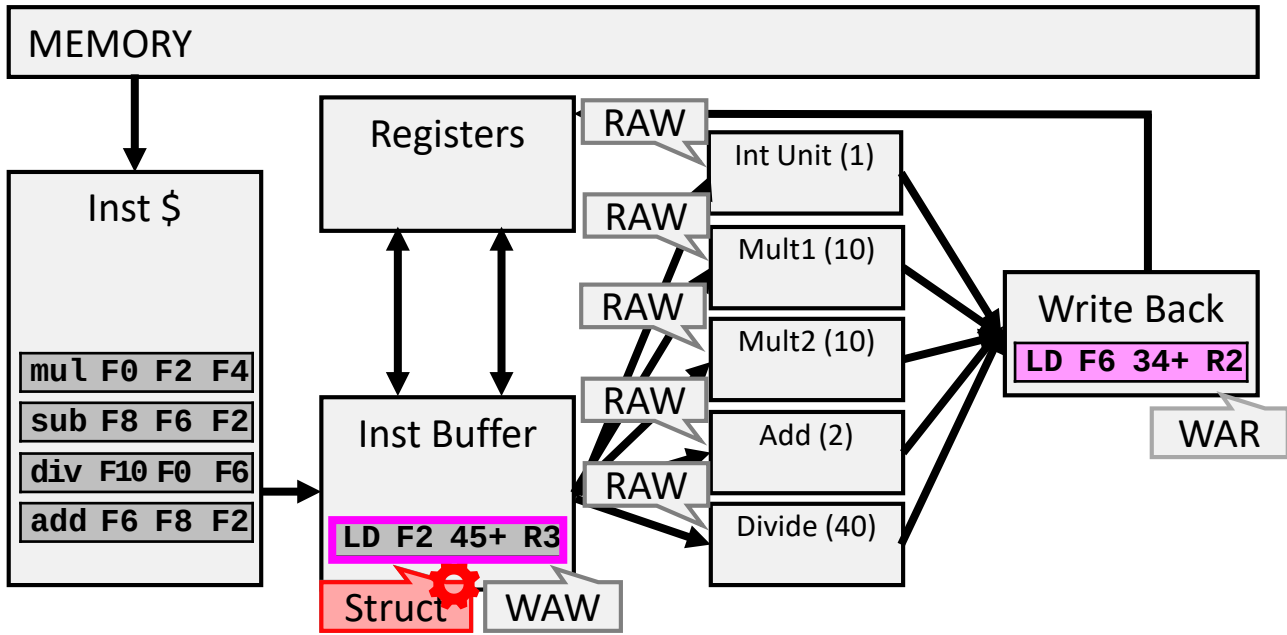
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	
LD	F2	45+	R3				
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F6		R2				yes
Mult1	No								
Mult2	No								
Add	No								
Div	No								

Reg Status	
	FU
F0	
F2	
F4	
F6	Int
F8	
F10	
...	

Scoreboard Example: Cycle 4



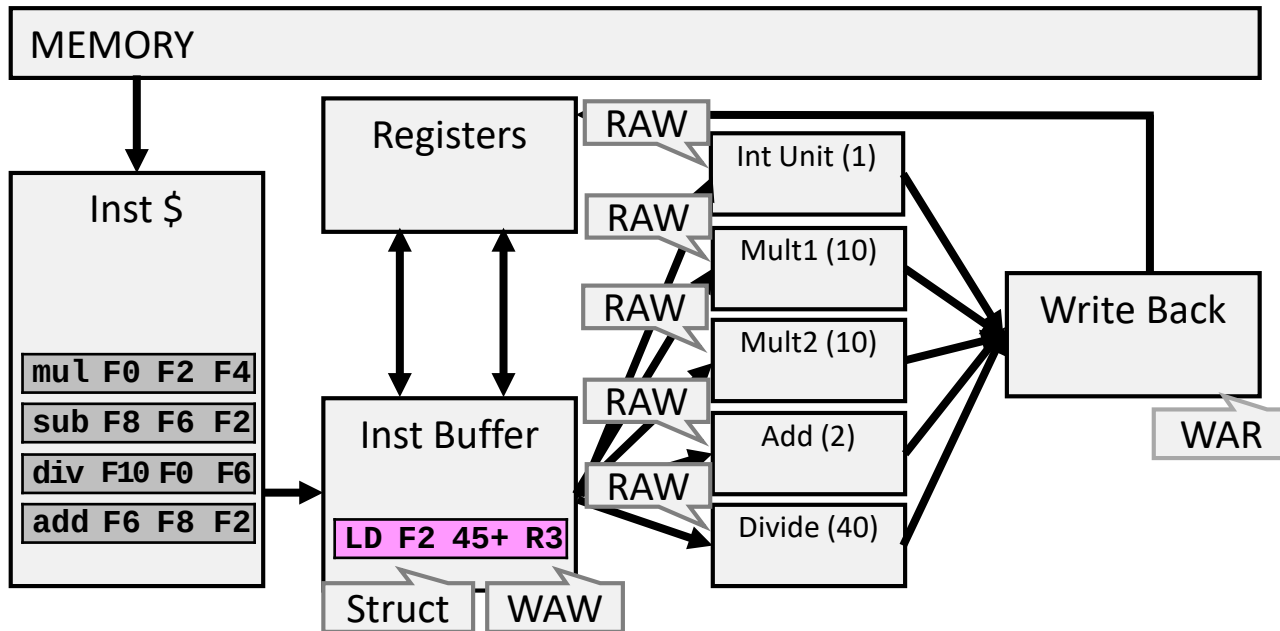
WB LD#1

Free FU & Reg status of LD#1

Integer unit free
→ No Hazard!

Instn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2		FU
LD	F6	34+	R2	1	2	3	4	Int	No									F0	
LD	F2	45+	R3					Mult1	No									F2	
MULTD	F0	F2	F4					Mult2	No									F4	
SUBD	F8	F6	F2					Add	No									F6	
DIVD	F10	F0	F6					Div	No									F8	
ADD	F6	F8	F2															F10	
																		...	

Scoreboard Example: **Cycle 5**



Issue LD #2 since
int unit is now free.

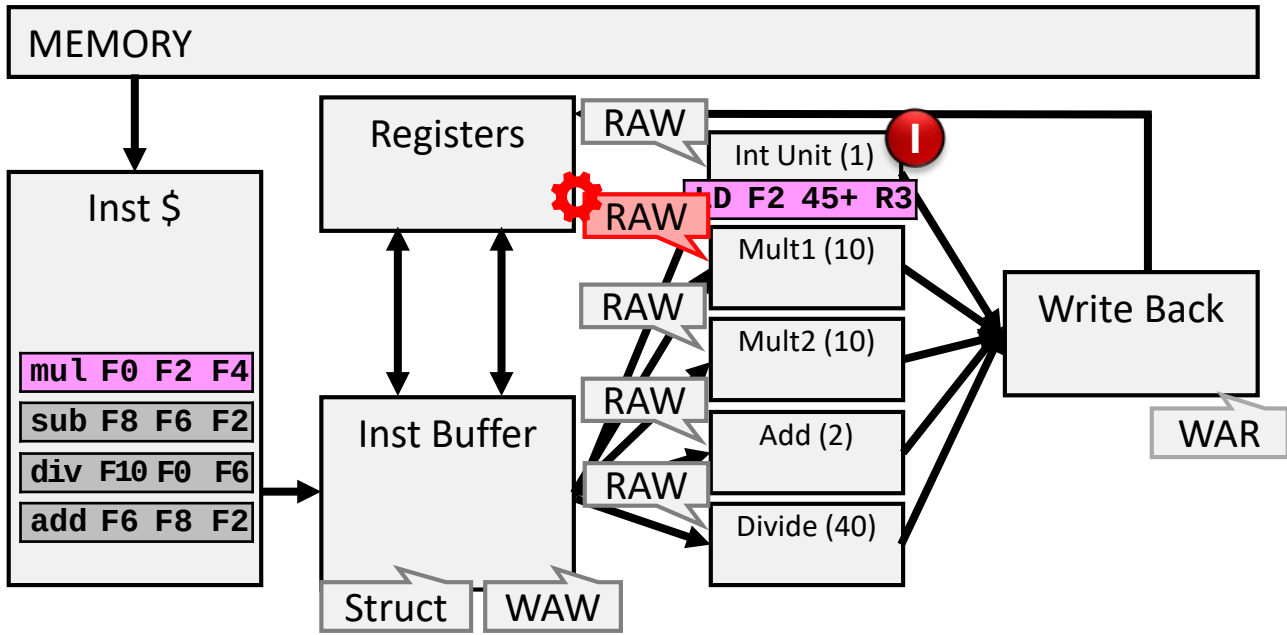
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5			
MULTD	F0	F2	F4				
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F2		R3				yes
Mult1	No								
Mult2	No								
Add	No								
Div	No								

Reg Status	
	FU
F0	
F2	Int
F4	
F6	
F8	
F10	
...	

Scoreboard Example: Cycle 6



Read operand
of LD#2

Try to Issue MULT

Check Hazards
→ No Hazards!

NOTE) There is
RAW due to LD#2

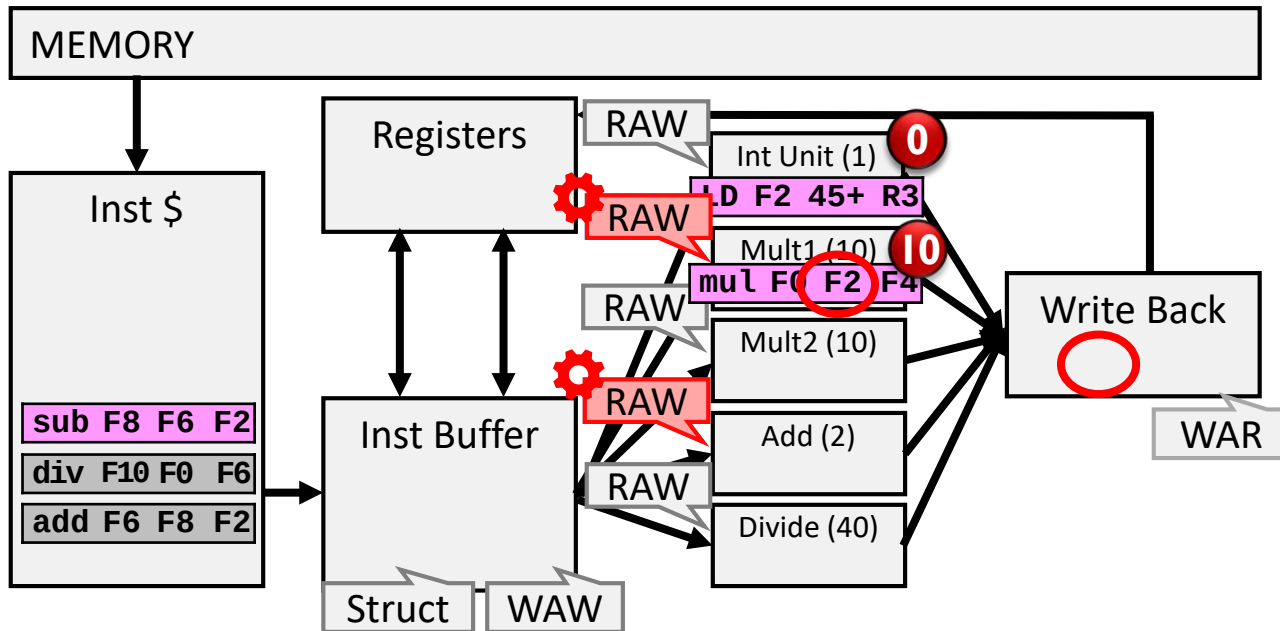
Scoreboard

Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6		
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2				
DIVD	F10	F0	F6				
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F2	R3					yes
Mult1	Yes	MUL	F0	F2	F4	Int		No	yes
Mult2	No								
Add	No								
Div	No								

Reg Status	
Reg	FU
F0	Mult1
F2	Int
F4	
F6	
F8	
F10	
...	

➤ Scoreboard Example: 🕒 Cycle 7



Execute LD#2

LD#2 comp. done

MULT can't read its
operands (F2) because
LD #2 hasn't finished.

Issue SUB

NOTE) There is RAW due to LD#2

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2	7			
DIVD	F10	F0	F6				
ADDD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	Yes	LD	F2		R3				yes
Mult1	Yes	MUL	F0	F2	F4	Int		No	yes
Mult2	No								
Add	Yes	SUB	F8	F6	F2		Int	Yes	No
Div	No								

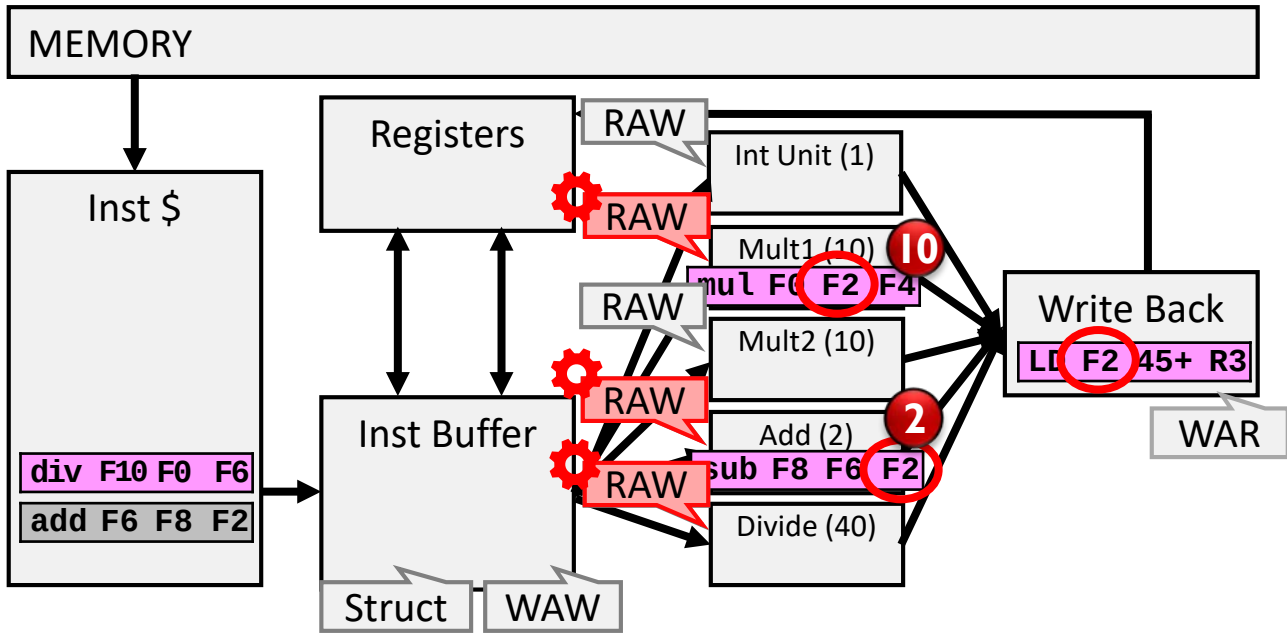
Reg Status	
	FU
F0	Mult1
F2	Int
F4	
F6	
F8	Add
F10	
...	

Scoreboard Example: Cycle 8a

First half of clock cycle

Issue DIVD

MULT & SUBD both waiting for F2

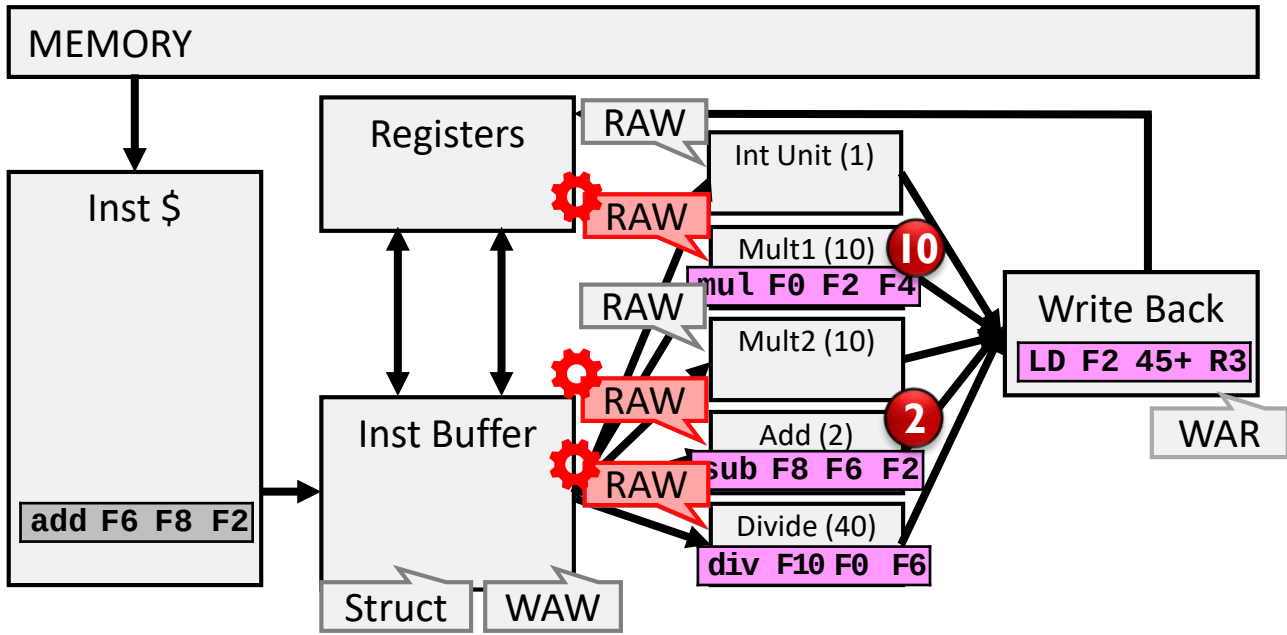


NOTE) There is RAW due to Mult1

Insn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2	Reg	FU
LD	F6	34+	R2	1	2	3	4	Int	Yes	LD	F2		R3				yes	F0	Mult1
LD	F2	45+	R3	5	6	7		Mult1	Yes	MUL	F0	F2	F4	Int		No	yes	F2	Int
MULTD	F0	F2	F4	6				Mult2	No									F4	
SUBD	F8	F6	F2	7				Add	Yes	SUB	F8	F6	F2		Int	Yes	No	F6	
DIVD	F10	F0	F6	8				Div	Yes	DIV	F10	F0	F6	Mult1		No	Yes	F8	Add
ADD	F6	F8	F2															F10	Div
																		...	

Scoreboard Example: Cycle 8b

Second half of clock cycle



LD#2 writes F2

Free FU & Reg status of LD#2

Now, F2 is ready

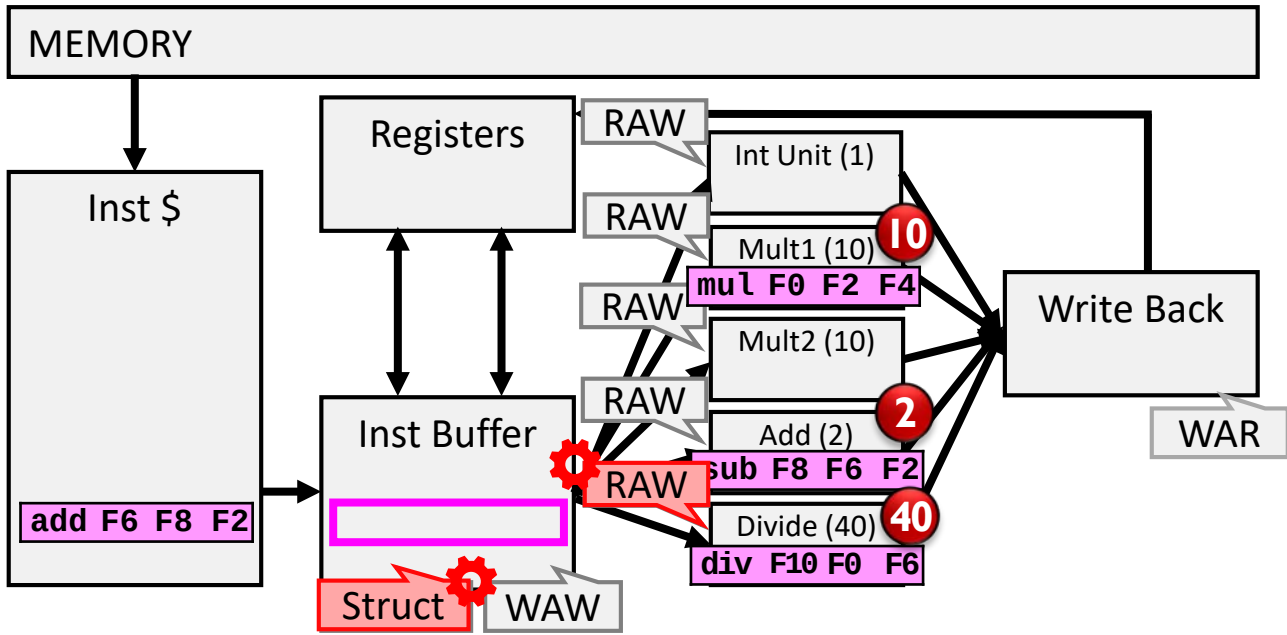
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6			
SUBD	F8	F6	F2	7			
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	SUB	F8	F6	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mult1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	
F8	Add
F10	Div
...	

Scoreboard Example: Cycle 9



Try to Issue MULT

ADDD cant start
∴ add unit is busy

Now MULT and SUBD
can both read F2.

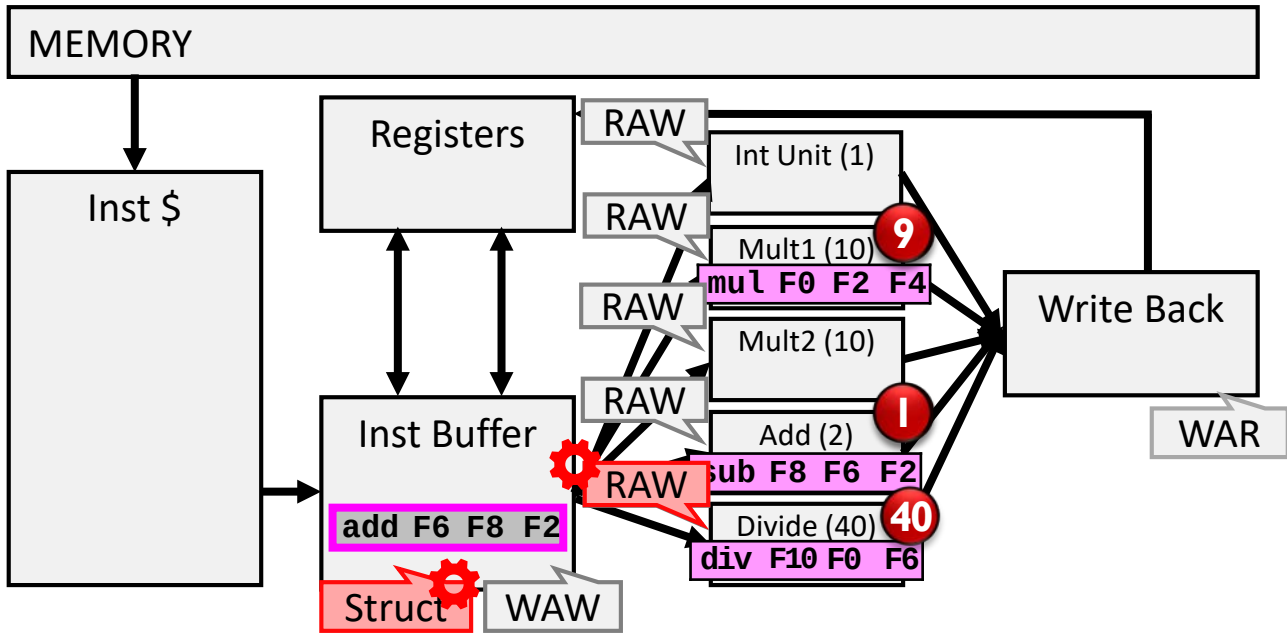
Scoreboard							
Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9		
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	SUB	F8	F6	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	
F8	Add
F10	Div
...	

Scoreboard Example: Cycle 10

Calculating....



Scoreboard

Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9		
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2				

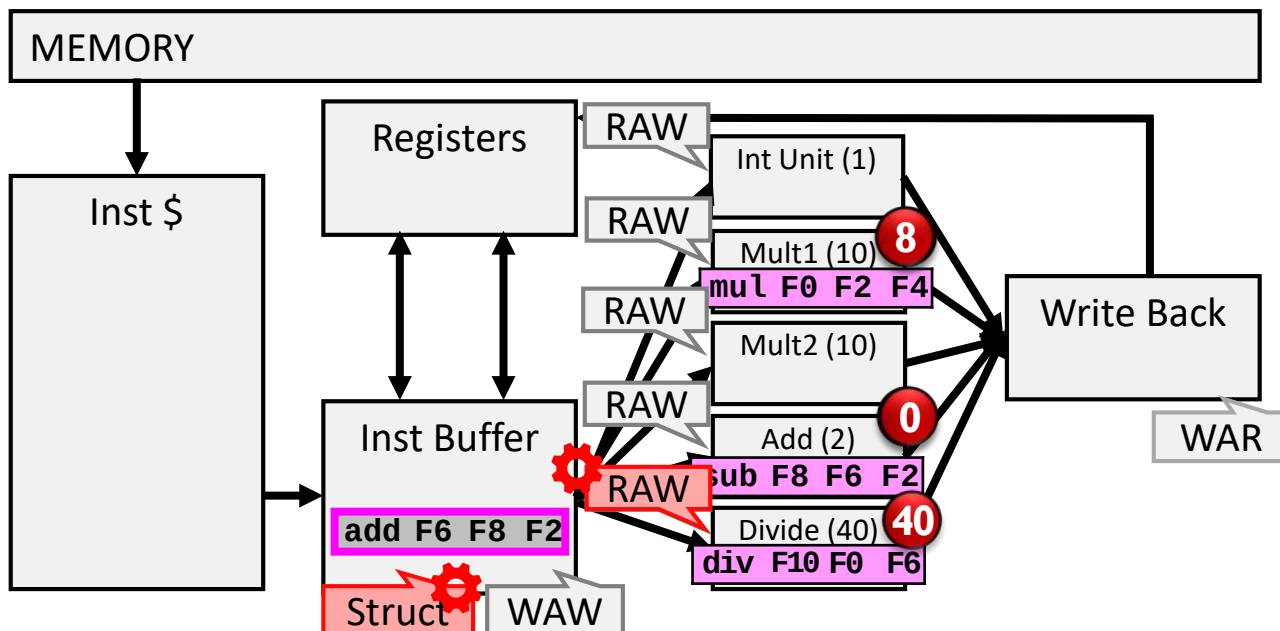
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	SUB	F8	F6	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	
F8	Add
F10	Div
...	

Scoreboard Example: Cycle 11

Calculating....

SUB comp. done



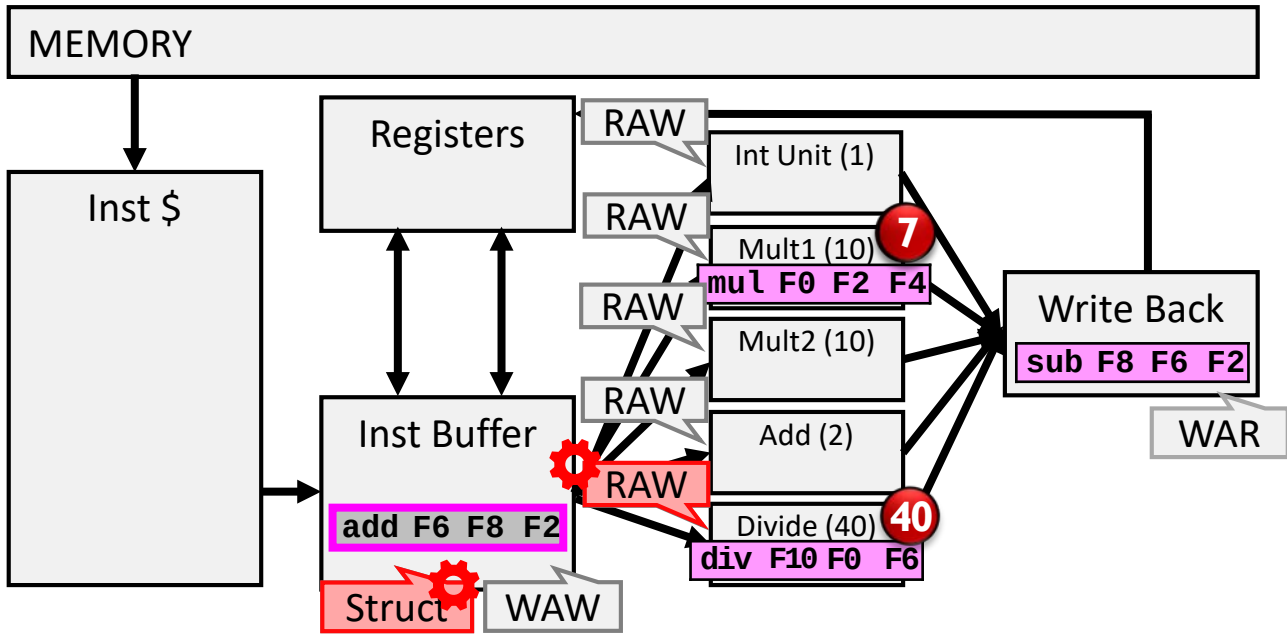
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2				

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	SUB	F8	F6	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	
F8	Add
F10	Div
...	

Scoreboard Example: Cycle 12



- Calculating....
- SUB writes F8
- Free FU & Reg status of SUB
- Integer unit free → No Hazard!

Scoreboard

Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2				

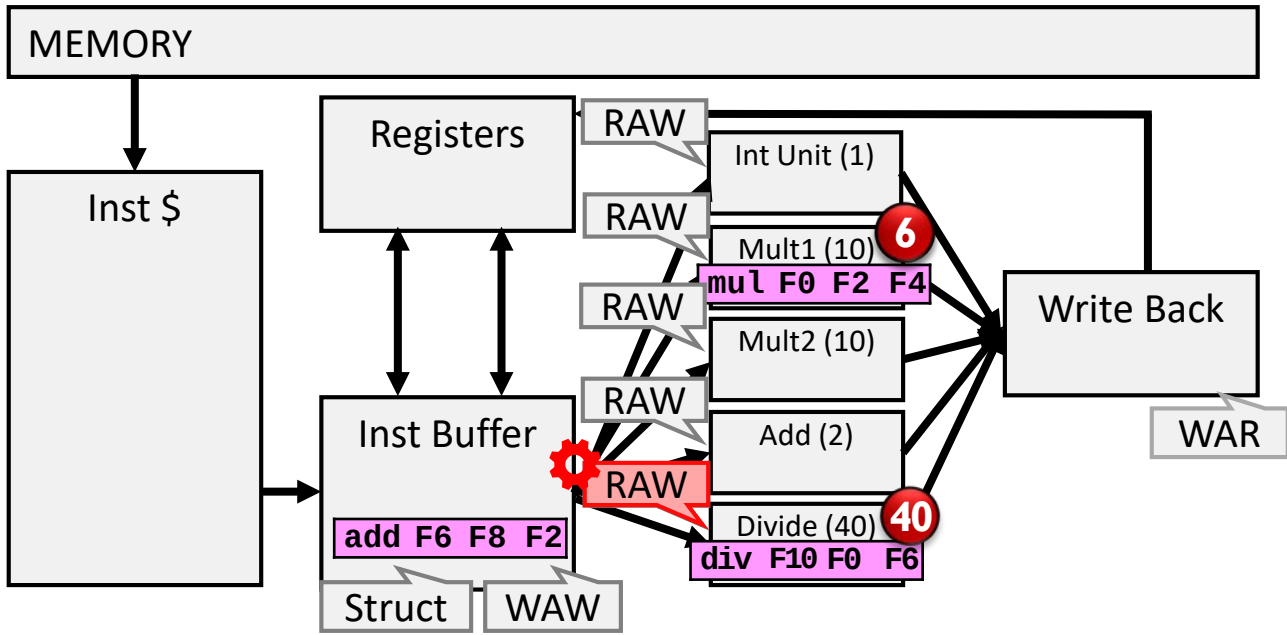
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	No								
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	
F8	
F10	Div
...	

Scoreboard Example: Cycle 13

Calculating....

Issue ADDD



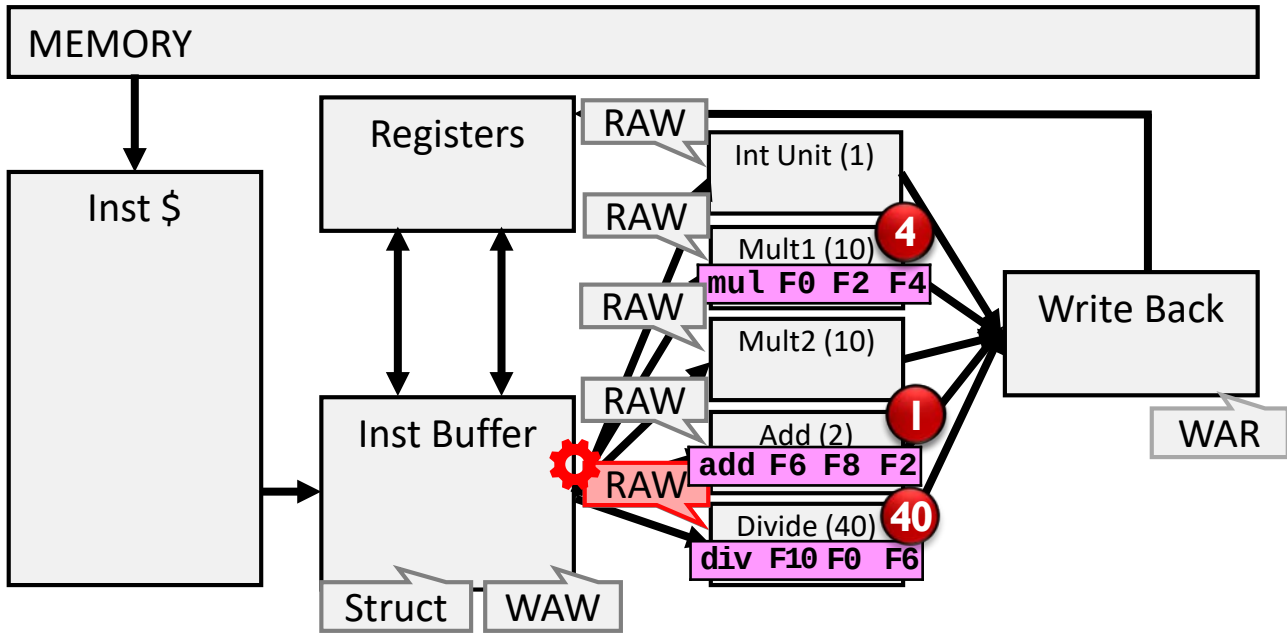
Scoreboard							
Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADDD	F6	F8	F2	13			

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 15

Calculating....



Scoreboard							
Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2	13	14		

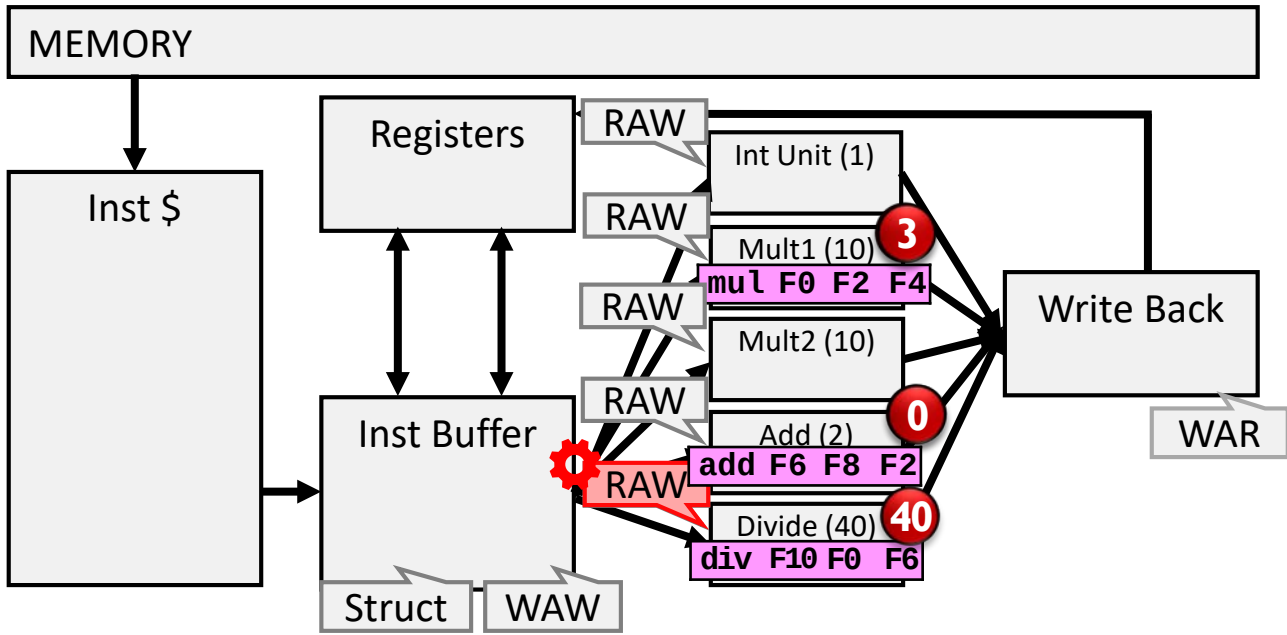
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 16

Calculating....

ADD comp. done



Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2	13	14	16	

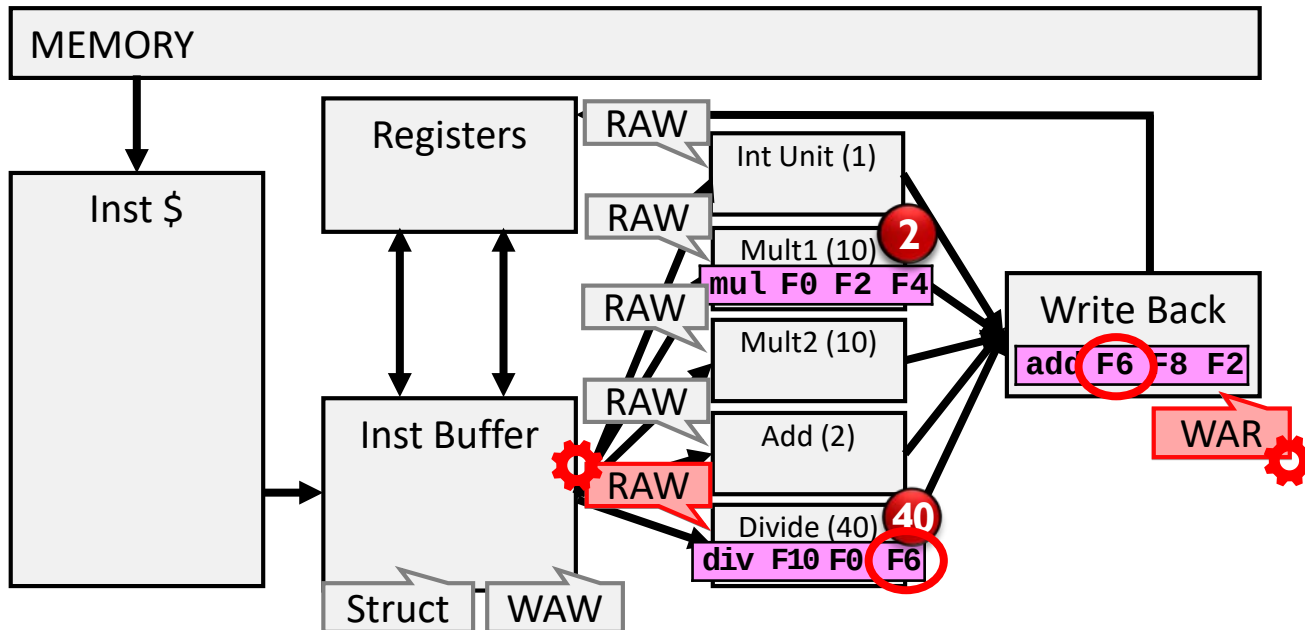
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 17

Calculating....

Oops! ADD
cant write
because of
DIVD. WAR
hazard!



Scoreboard

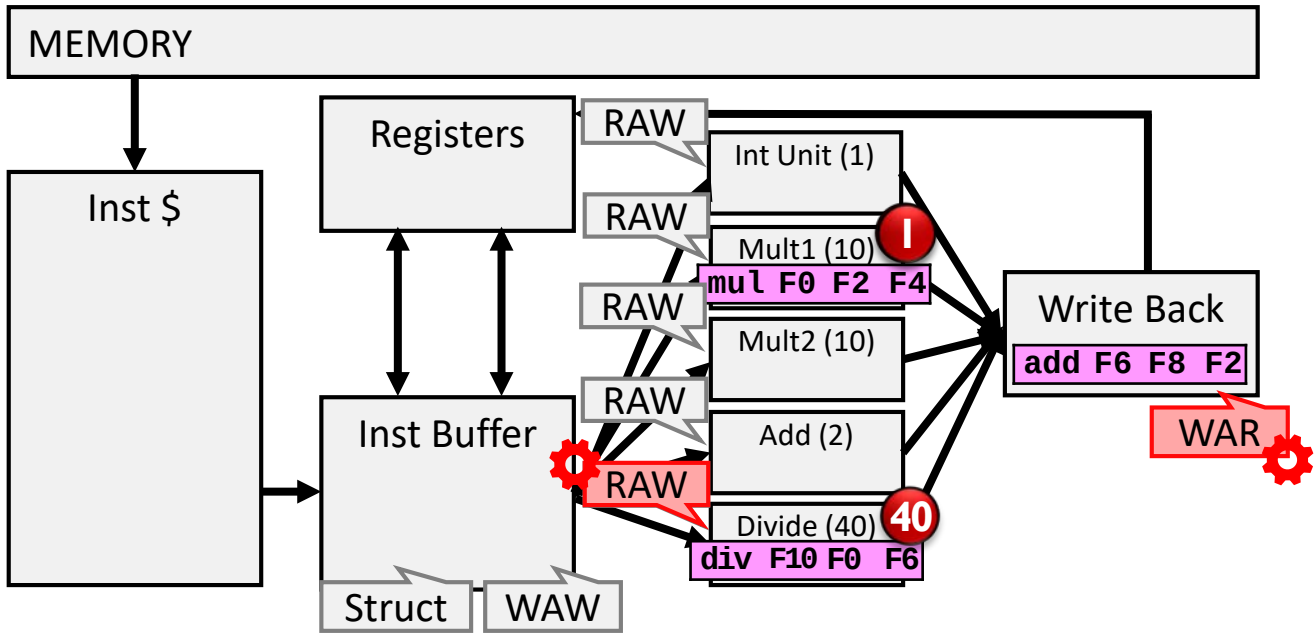
Instn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9		
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2	13	14	16	

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	Yes	MUL	F0	F2	F4			Yes	yes
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		No	Yes

Reg Status	
	FU
F0	Mult1
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 18

Calculating....

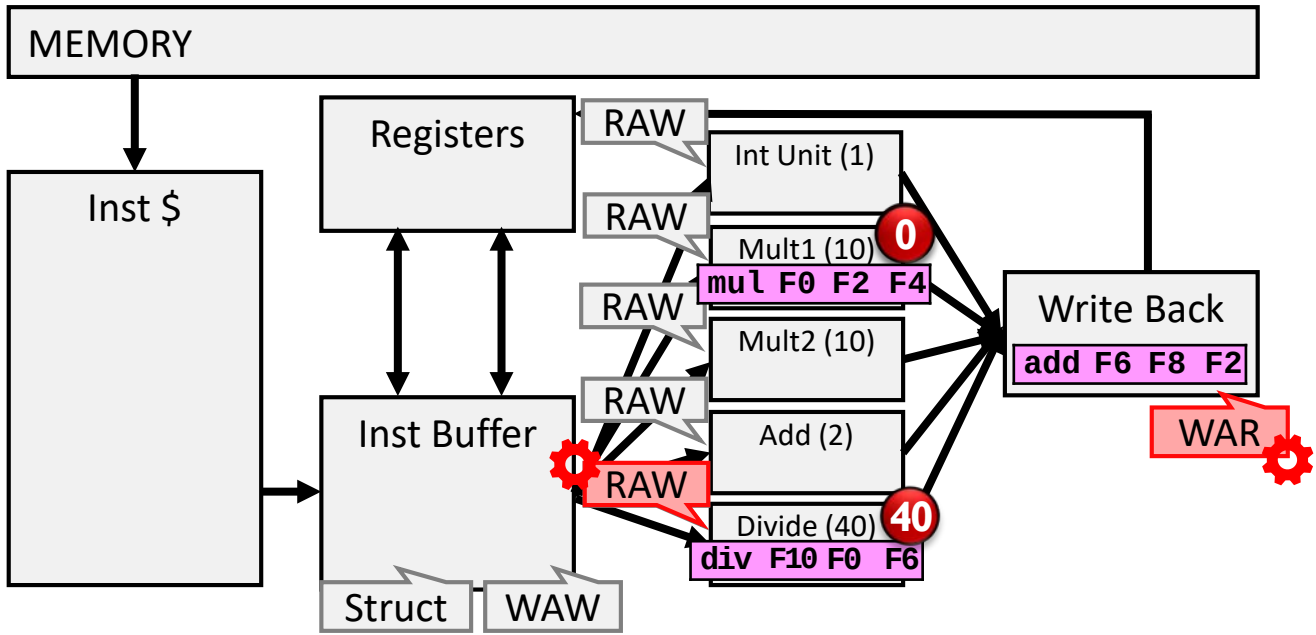


Instn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2	Reg	FU
LD	F6	34+	R2	1	2	3	4	Int	No									F0	Mult1
LD	F2	45+	R3	5	6	7	8	Mult1	Yes	MUL	F0	F2	F4			Yes	yes	F2	
MULTD	F0	F2	F4	6	9			Mult2	No									F4	
SUBD	F8	F6	F2	7	9	11	12	Add	Yes	ADD	F6	F8	F2			Yes	Yes	F6	Add
DIVD	F10	F0	F6	8				Div	Yes	DIV	F10	F0	F6	Mult1		No	Yes	F8	
ADD	F6	F8	F2	13	14	16												F10	Div
																		...	

Scoreboard Example: Cycle 19

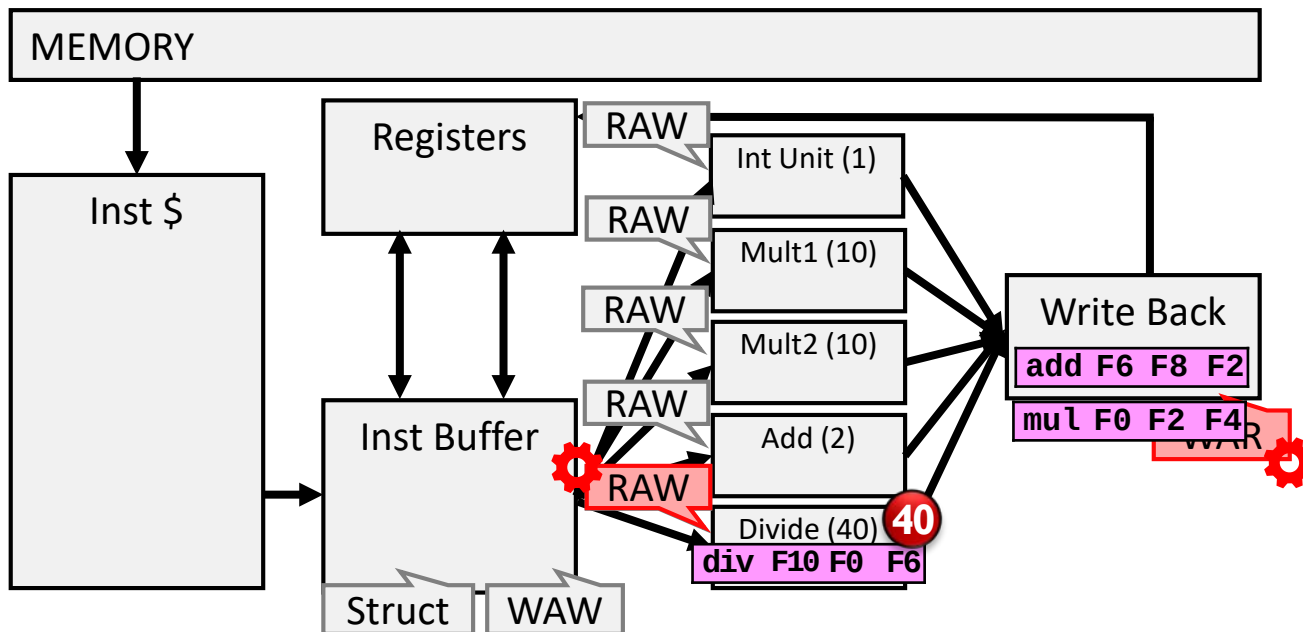
Calculating....

MULTD comp. done



Instn Status								FU Status										Reg Status	
Inst	dst	src1	src2	D	S	X	W	FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2	Reg	FU
LD	F6	34+	R2	1	2	3	4	Int	No									F0	Mult1
LD	F2	45+	R3	5	6	7	8	Mult1	Yes	MUL	F0	F2	F4			Yes	yes	F2	
MULTD	F0	F2	F4	6	9	19		Mult2	No									F4	
SUBD	F8	F6	F2	7	9	11	12	Add	Yes	ADD	F6	F8	F2			Yes	Yes	F6	Add
DIVD	F10	F0	F6	8				Div	Yes	DIV	F10	F0	F6	Mult1		No	Yes	F8	
ADD	F6	F8	F2	13	14	16												F10	Div
																		...	

Scoreboard Example: Cycle 20



MULTD writes F0

Free FU & Reg status of MULTD

Now, F0 is ready

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8			
ADD	F6	F8	F2	13	14	16	

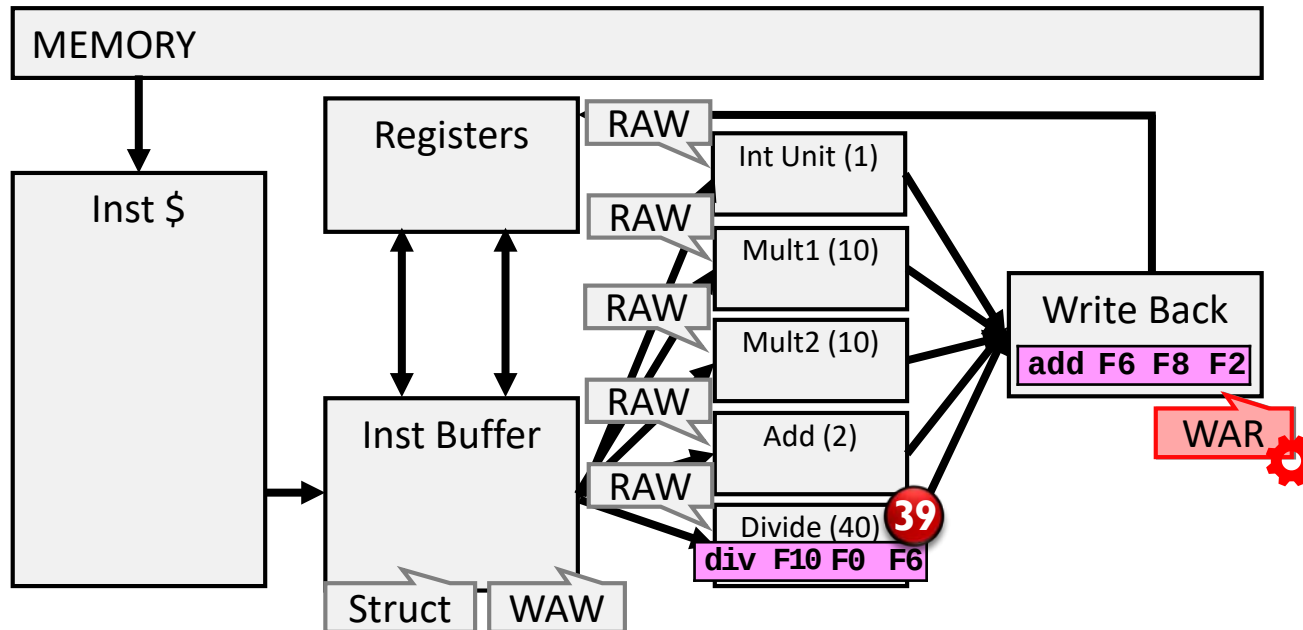
FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	No								
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6			Yes	Yes

Reg Status	
	FU
F0	
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 21

DIVD loads operands

WAR is removed!



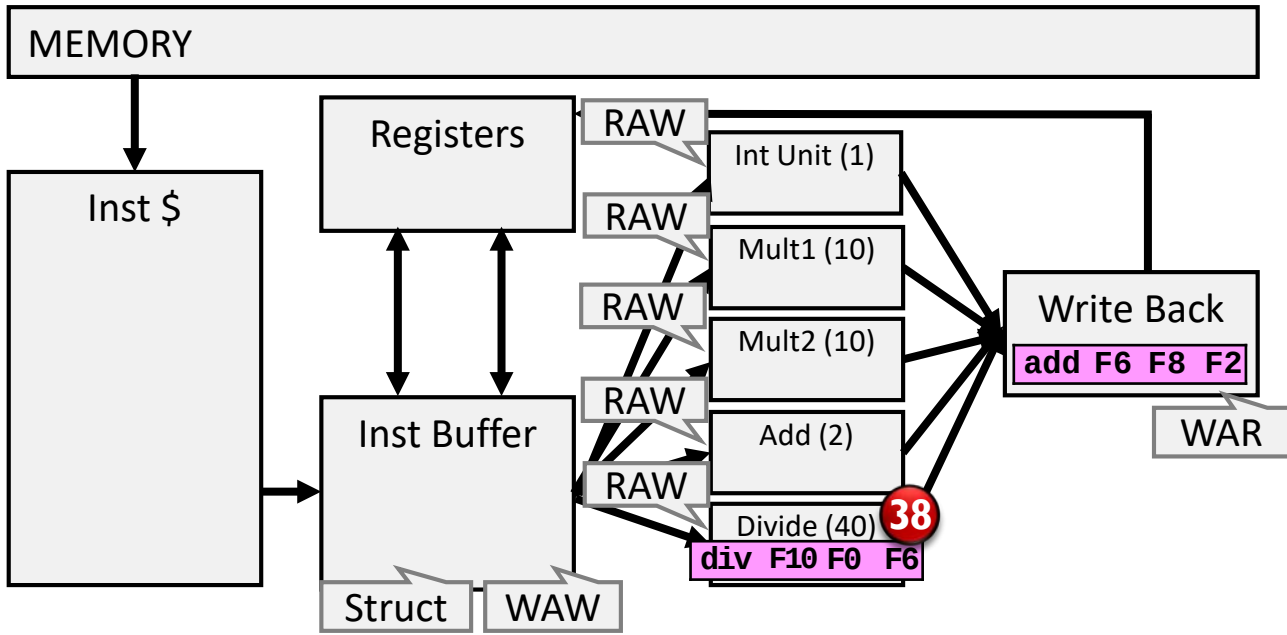
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21		
ADD	F6	F8	F2	13	14	16	

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	No								
Mult2	No								
Add	Yes	ADD	F6	F8	F2			Yes	Yes
Div	Yes	DIV	F10	F0	F6	Mul1		Yes	Yes

Reg Status	
	FU
F0	
F2	
F4	
F6	Add
F8	
F10	Div
...	

Scoreboard Example: Cycle 22



Calculating....

ADDD writes F6

Free FU & Reg status of MULTD

Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21		
ADDD	F6	F8	F2	13	14	16	22

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	No								
Mult2	No								
Add	No								
Div	Yes	DIV	F10	F0	F6	Mul1		Yes	Yes

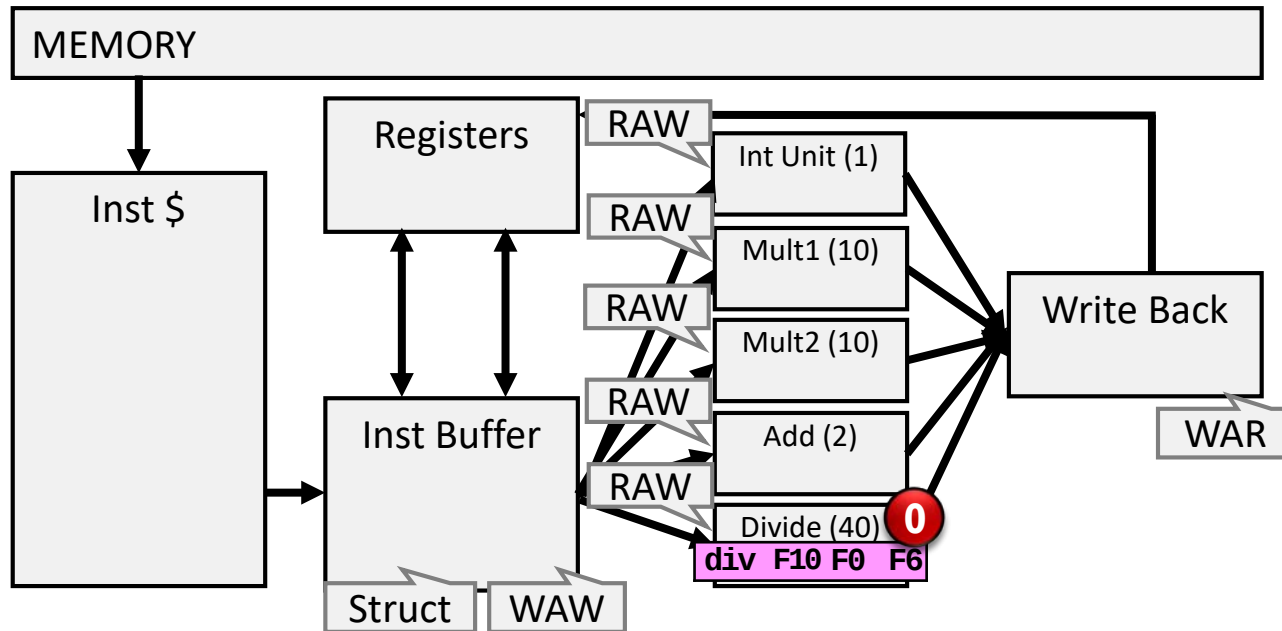
Reg Status	
	FU
F0	
F2	
F4	
F6	
F8	
F10	Div
...	

**Faster than light computation
(skip a couple of cycles)**

Scoreboard Example: Cycle 61

Calculating....

DIVD comp. done



Scoreboard

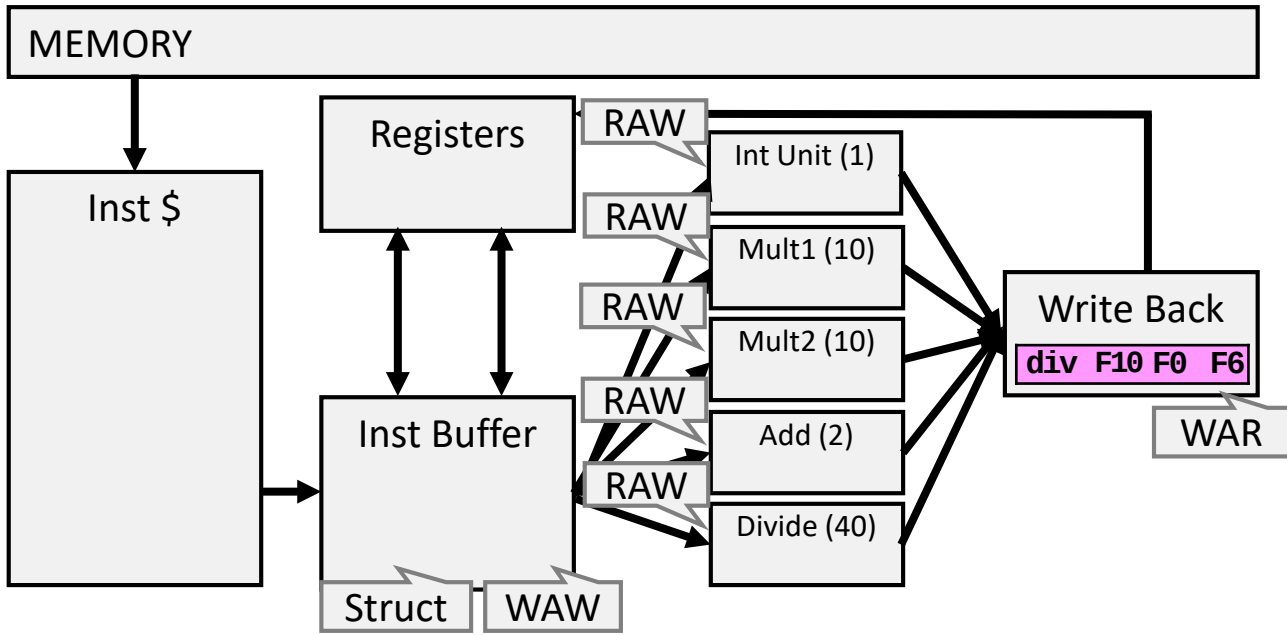
Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21	61	
ADD	F6	F8	F2	13	14	16	22

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	No								
Mult2	No								
Add	No								
Div	Yes	DIV	F10	F0	F6	Mul1		Yes	Yes

Reg Status	
	FU
F0	
F2	
F4	
F6	
F8	
F10	Div
...	

Scoreboard Example: Cycle 62

DONE!!!!



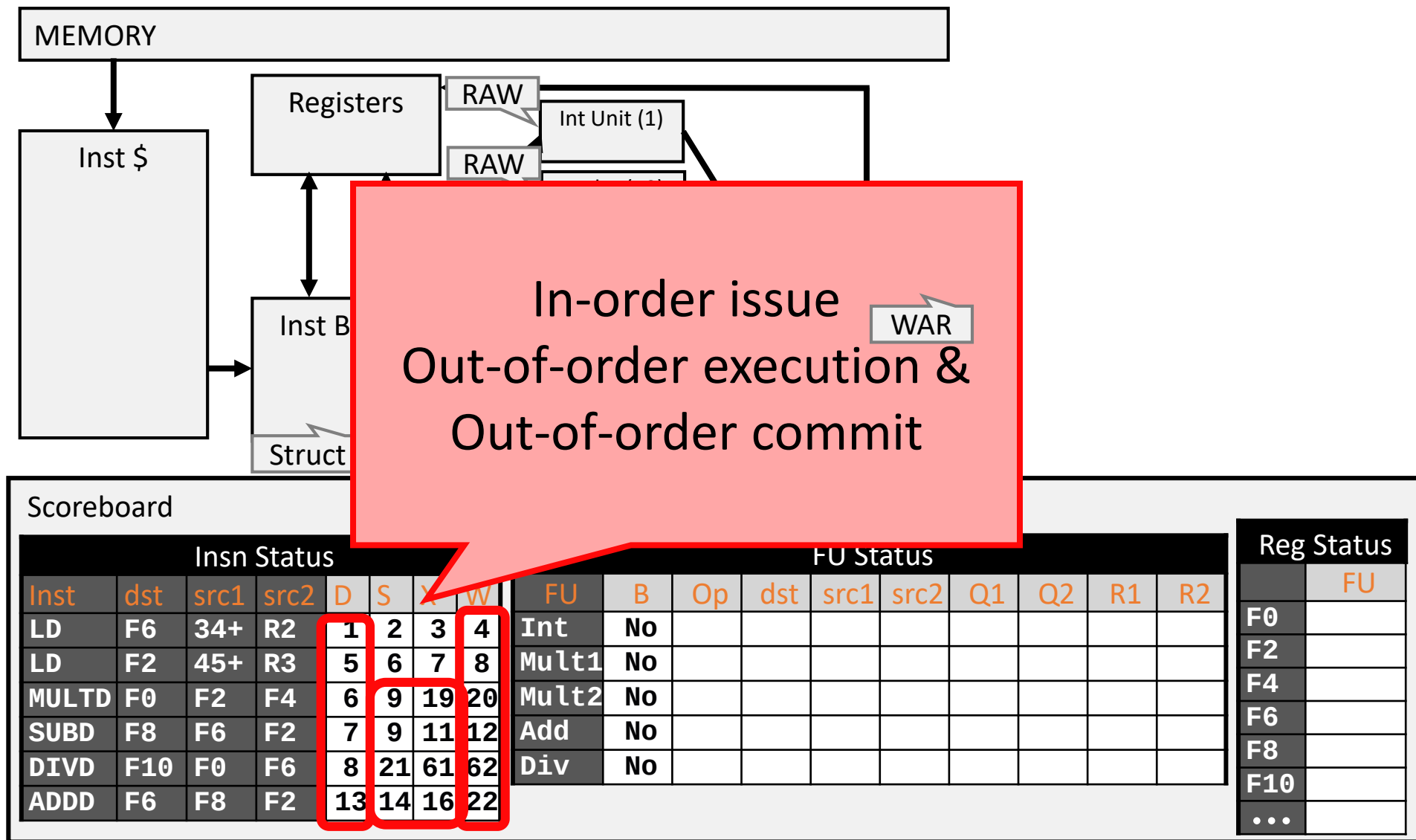
Scoreboard

Insn Status							
Inst	dst	src1	src2	D	S	X	W
LD	F6	34+	R2	1	2	3	4
LD	F2	45+	R3	5	6	7	8
MULTD	F0	F2	F4	6	9	19	20
SUBD	F8	F6	F2	7	9	11	12
DIVD	F10	F0	F6	8	21	61	62
ADD	F6	F8	F2	13	14	16	22

FU Status									
FU	B	Op	dst	src1	src2	Q1	Q2	R1	R2
Int	No								
Mult1	No								
Mult2	No								
Add	No								
Div	No								

Reg Status	
	FU
F0	
F2	
F4	
F6	
F8	
F10	
...	

Review of Cycle 62



Scoreboard Summary

- **The good**

- + Cheap hardware

- * InsnStatus + FuStatus + RegStatus ~ 1FP unit in area

- + Pretty good performance

- * 1.7X for FORTRAN (scientific array) programs

- **The less good**

- No bypassing

- * Is this a fundamental problem?

- Limited scheduling scope

- * Structural/**WAW** hazards delay dispatch

- Slow issue of truly-dependent (RAW) instructions

- * WAR hazards delay writeback

- Fix with **hardware register renaming**

Dynamic Scheduling - Scoreboard -

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